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Inventor(s)	Ito; Masahiro et al.

Heterocyclic compound

Abstract

The present invention provides a heterocyclic compound having a HDAC6 inhibitory action, which is useful for the treatment of central nervous system diseases including neurodegenerative diseases, and the like, and a medicament comprising the compound. The present invention relates to a compound represented by the formula (I): ##STR00001## wherein each symbol is as defined in the description, or a salt thereof.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7250504	12/2006	Grozinger et al.	N/A	N/A
7879865	12/2010	Richon et al.	N/A	N/A
8026260	12/2010	Close et al.	N/A	N/A
8067472	12/2010	Richon et al.	N/A	N/A
8076116	12/2010	Grozinger et al.	N/A	N/A
8088951	12/2011	Tsai et al.	N/A	N/A
8389553	12/2012	Harrington et al.	N/A	N/A
8435780	12/2012	Grozinger et al.	N/A	N/A
8895284	12/2013	Grozinger et al.	N/A	N/A
8901156	12/2013	Baloglu et al.	N/A	N/A
8981084	12/2014	Baloglu et al.	N/A	N/A
8999289	12/2014	Anderson et al.	N/A	N/A
9056843	12/2014	Hebach et al.	N/A	N/A
9096559	12/2014	Harrington et al.	N/A	N/A
9238028	12/2015	Van Den Bosch et al.	N/A	N/A
9365498	12/2015	Holson et al.	N/A	N/A
9447030	12/2015	Holson et al.	N/A	N/A
9480673	12/2015	Freed et al.	N/A	N/A
9512083	12/2015	Raje et al.	N/A	N/A
9572854	12/2016	Anderson et al.	N/A	N/A
9670193	12/2016	Hebach et al.	N/A	N/A
9890172	12/2017	Holson et al.	N/A	N/A
10130604	12/2017	Freed et al.	N/A	N/A
10172905	12/2018	Anderson et al.	N/A	N/A
10406146	12/2018	Kaieda et al.	N/A	N/A
10584117	12/2019	Lee et al.	N/A	N/A

11453661	12/2021	Ito	N/A	C07D 413/10
2003/0129724	12/2002	Grozinger et al.	N/A	N/A
2004/0077591	12/2003	Dangond	N/A	N/A
2004/0087657	12/2003	Richon et al.	N/A	N/A
2004/0142859	12/2003	Steffan et al.	N/A	N/A
2005/0009030	12/2004	Schweighoffer et al.	N/A	N/A
2005/0227915	12/2004	Steffan et al.	N/A	N/A
2006/0018921	12/2005	Levenson et al.	N/A	N/A
2006/0135612	12/2005	Ferrante	N/A	N/A
2007/0078083	12/2006	Barlow et al.	N/A	N/A
2007/0207950	12/2006	Yao et al.	N/A	N/A
2008/0114069	12/2007	Richon et al.	N/A	N/A
2008/0221192	12/2007	Wan et al.	N/A	N/A
2009/0036318	12/2008	Grozinger et al.	N/A	N/A
2009/0136461	12/2008	Kim et al.	N/A	N/A
2010/0004156	12/2009	Kaushal et al.	N/A	N/A
2010/0022514	12/2009	Cho et al.	N/A	N/A
2010/0247543	12/2009	Maes et al.	N/A	N/A
2010/0292169	12/2009	Yao et al.	N/A	N/A
2011/0124731	12/2010	Richon et al.	N/A	N/A
2012/0039909	12/2011	Tsai et al.	N/A	N/A
2012/0041067	12/2011	Richon et al.	N/A	N/A
2012/0289495	12/2011	Baloglu et al.	N/A	N/A
2013/0059883	12/2012	Baloglu et al.	N/A	N/A
2013/0116204	12/2012	Schafer et al.	N/A	N/A
2013/0203735	12/2012	Sisko et al.	N/A	N/A
2013/0225543	12/2012	Jones et al.	N/A	N/A
2013/0227717	12/2012	Van Den Bosch et al.	N/A	N/A
2013/0317003	12/2012	Jacques et al.	N/A	N/A
2014/0051680	12/2013	Jacques et al.	N/A	N/A
2014/0080800	12/2013	Holson et al.	N/A	N/A
2014/0080802	12/2013	Holson et al.	N/A	N/A
2014/0235649	12/2013	Kovach et al.	N/A	N/A
2014/0329825	12/2013	Hebach et al.	N/A	N/A
2014/0378385	12/2013	Raje et al.	N/A	N/A
2015/0038534	12/2014	Baloglu et al.	N/A	N/A
2015/0119327	12/2014	Muotri et al.	N/A	N/A
2015/0158884	12/2014	Jacques et al.	N/A	N/A
2015/0184154	12/2014	Matthias et al.	N/A	N/A
2016/0000766	12/2015	Van Den Bosch et al.	N/A	N/A
2016/0221972	12/2015	Zheng et al.	N/A	N/A
2016/0221973	12/2015	Zheng et al.	N/A	N/A
2016/0222022	12/2015	Zheng et al.	N/A	N/A
2016/0222028	12/2015	Zheng et al.	N/A	N/A
2016/0251351	12/2015	Holson et al.	N/A	N/A
2016/0347761	12/2015	Holson et al.	N/A	N/A

2017/0015655	12/2016	Kaieda et al.	N/A	N/A
2017/0044186	12/2016	Jacques et al.	N/A	N/A
2017/0050984	12/2016	Jacques et al.	N/A	N/A
2017/0096405	12/2016	Song et al.	N/A	N/A
2017/0217955	12/2016	Holson et al.	N/A	N/A
2017/0305866	12/2016	Raje et al.	N/A	N/A
2018/0016282	12/2017	Holson et al.	N/A	N/A
2018/0016314	12/2017	Harley et al.	N/A	N/A
2018/0099977	12/2017	Holson et al.	N/A	N/A
2018/0208606	12/2017	Jacques et al.	N/A	N/A
2018/0230113	12/2017	Lee et al.	N/A	N/A
2018/0256572	12/2017	Yates	N/A	N/A
2018/0273495	12/2017	Kim et al.	N/A	N/A
2018/0327423	12/2017	Jacques et al.	N/A	N/A
2018/0370987	12/2017	Jacques et al.	N/A	N/A
2019/0016692	12/2018	Zheng et al.	N/A	N/A
2019/0016693	12/2018	Zheng et al.	N/A	N/A
2019/0038589	12/2018	Freed et al.	N/A	N/A
2019/0062288	12/2018	Zheng et al.	N/A	N/A
2019/0112279	12/2018	Zheng et al.	N/A	N/A
2019/0112280	12/2018	Zheng et al.	N/A	N/A
2019/0112281	12/2018	Zheng et al.	N/A	N/A
2019/0112282	12/2018	Zheng et al.	N/A	N/A
2019/0112283	12/2018	Zheng et al.	N/A	N/A
2019/0112284	12/2018	Zheng et al.	N/A	N/A
2019/0119227	12/2018	Zheng et al.	N/A	N/A
2019/0119228	12/2018	Zheng et al.	N/A	N/A
2019/0119229	12/2018	Zheng et al.	N/A	N/A
2019/0119230	12/2018	Zheng et al.	N/A	N/A
2019/0119231	12/2018	Zheng et al.	N/A	N/A
2019/0119232	12/2018	Zheng et al.	N/A	N/A
2019/0119233	12/2018	Zheng et al.	N/A	N/A
2019/0119234	12/2018	Zheng et al.	N/A	N/A
2019/0119235	12/2018	Zheng et al.	N/A	N/A
2019/0127339	12/2018	Zheng et al.	N/A	N/A
2019/0135799	12/2018	Ito et al.	N/A	N/A
2019/0233439	12/2018	Jacques et al.	N/A	N/A
2020/0002298	12/2019	Zheng et al.	N/A	N/A
2020/0048211	12/2019	Zheng et al.	N/A	N/A
2020/0048212	12/2019	Zheng et al.	N/A	N/A
2020/0055830	12/2019	Zheng et al.	N/A	N/A
2020/0148704	12/2019	Jacques et al.	N/A	N/A
2020/0354338	12/2019	Jin et al.	N/A	N/A
2021/0009538	12/2020	Zheng et al.	N/A	N/A
2021/0009539	12/2020	Zheng et al.	N/A	N/A
2021/0009540	12/2020	Zheng et al.	N/A	N/A
2022/0204461	12/2021	Zheng et al.	N/A	N/A
2022/0235035	12/2021	Jin et al.	N/A	N/A
2022/0363651	12/2021	Zheng et al.	N/A	N/A
2023/0021554	12/2022	Jin et al.	N/A	N/A

2023/0023761	12/2022	Jin et al.	N/A	N/A
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FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2013203735	12/2012	AU	N/A
2016338118	12/2016	AU	N/A
101056632	12/2006	CN	N/A
102770412	12/2011	CN	N/A
105120860	12/2014	CN	N/A
106232584	12/2015	CN	N/A
107873023	12/2017	CN	N/A
107980040	12/2017	CN	N/A
108026088	12/2017	CN	N/A
1 390 491	12/2003	EP	N/A
1 746 982	12/2006	EP	N/A
2 624 832	12/2012	EP	N/A
3 187 497	12/2016	EP	N/A
3 327 019	12/2017	EP	N/A
WO-2005/105066	12/2004	WO	N/A
WO-2007/049262	12/2006	WO	N/A
WO-2007/147868	12/2006	WO	N/A
WO-2008/073733	12/2007	WO	N/A
WO-2008/126932	12/2007	WO	N/A
WO-2013/006408	12/2012	WO	N/A
WO-2013/009810	12/2012	WO	N/A
WO-2013/009827	12/2012	WO	N/A
WO-2013/009830	12/2012	WO	N/A
WO-2013/066831	12/2012	WO	N/A
WO-2013/066833	12/2012	WO	N/A
WO-2013/066835	12/2012	WO	N/A
WO-2013/066838	12/2012	WO	N/A
WO-2013/066839	12/2012	WO	N/A
WO-2015/131788	12/2014	WO	N/A
WO-2016/039398	12/2015	WO	N/A
WO-2017/018803	12/2016	WO	N/A
WO-2017/018804	12/2016	WO	N/A
WO-2017/018805	12/2016	WO	N/A
WO-2017/023133	12/2016	WO	N/A
WO-2017/033946	12/2016	WO	N/A
WO-2017/222952	12/2016	WO	N/A
WO-2018/165520	12/2017	WO	N/A
WO-2018/215597	12/2017	WO	N/A
WO-2019/027054	12/2018	WO	N/A
WO-2019/030692	12/2018	WO	N/A
WO-2019/032652	12/2018	WO	N/A
WO-2020/022794	12/2019	WO	N/A
WO-2020/158762	12/2019	WO	N/A

OTHER PUBLICATIONS

Azad et al., "The future of epigenetic therapy in solid tumours-lessons from the past," *Nat. Rev. Clin. Oncol.*, Apr. 2, 2013, 10:256-266. cited by applicant

Chuang et al., "Multiple roles of HDAC inhibition in neurodegenerative conditions," *Trends in Neurosciences*, 2009, 32(11):591-601. cited by applicant

Chung et al., "A Therapeutic Strategy Uses Histone Deacetylase Inhibitors to Modulate the Expression of Genes Involved in the Pathogenesis of Rheumatoid Arthritis," *Molecular Therapy*, Nov. 2003, 8(5):707-717. cited by applicant

De Zoeten et al., "Histone Deacetylase 6 and Heat Shock Protein 90 Control the Functions of Foxp3 T-Regulatory Cells," *Molecular and Cellular Biology*, May 2011, 31(10):2066-2078. cited by applicant

Dubey et al., "Neurodegeneration and microtubule dynamics: death by a thousand cuts," *Frontiers in Cellular Neuroscience*, Sep. 9, 2015, 9(343):1-15. cited by applicant

Glauben et al., "Histone Hyperacetylation is Associated with Amelioration of Experimental Colitis in Mice," *The Journal of Immunology*, 2006, 176:5015-5022. cited by applicant

Goedert et al., "Frontotemporal Dementia: Implications for Understanding Alzheimer Disease," *Cold Spring Harbor Perspectives in Medicine*, 2012 (online Nov. 22, 2011), 4:a006254, 1-21. cited by applicant

Gouindarajan et al., "Reducing HDAC6 ameliorates cognitive deficits in a mouse model for Alzheimer's disease," *EMBO Mol. Med.*, 2013, 5:52-63. cited by applicant

Haberland et al., "The many roles of histone deacetylases in development and physiology: implications for disease and therapy," *Nature Reviews Genetics*, Jan. 2009, 10:32-42. cited by applicant

Hancock et al., "HDAC inhibitor therapy in autoimmunity and transplantation," *Ann. Rheum. Dis.*, 2012, 71(Supp.II):i46-i54. cited by applicant

Hubbert et al., "HDAC6 is a microtubule-associated deacetylase," *Nature*, May 23, 2002, 417:455-458. cited by applicant

Jin et al., "Design, synthesis and preliminary biological evaluation of indoline-2,3-dione derivatives as novel HDAC inhibitors," *Bioorganic & Medicinal Chemistry*, 2015, 23:4728-4736. cited by applicant

Jochems et al., "Antidepressant-Like Properties of Novel HDAC6-Selective Inhibitors with Improved Brain Bioavailability," *Neuropsychopharmacology*, 2014, 39:389-400. cited by applicant

Kalin et al., "Development and Therapeutic Implications of Selective Histone Deacetylase 6 Inhibitors," *Journal of Medicinal Chemistry*, Apr. 29, 2013, 56:6297-6313. cited by applicant

Leroux, Michel R., "Tubulin acetyltransferase discovered: Ciliary role in the ancestral eukaryote expanded to neurons in metazoans," *PNAS*, Dec. 14, 2020, 107(50):21238-21239. cited by applicant

Li et al., "HDAC inhibitor reduces cytokine storm and facilitates induction of chimerism that reverses lupus in anti-CD3 conditioning regimen," *PNAS*, Mar. 25, 2008, 105(12):4796-4801. cited by applicant

Ligand-Protein Interactions and Molecular Similarity, H. Kubinyi, Ed., Springer, 1998, vol. 2-3, 243-244. cited by applicant

Lin et al., "Anti-rheumatic activities of histone deacetylase (HDAC) inhibitors in vivo in collagen-induced arthritis in rodents," *British Journal of Pharmacology*, 2007, 150:862-872. cited by applicant

Liu et al., "Histone deacetylase 6: structure, functions and development of selective inhibitors," *Acta Pharmaceutica Sinica*, 2015, 50(1):7-14. cited by applicant

Lu et al., *Journal of International Neurology and Neurosurgery*, Dec. 31, 2011, 348-352, with English abstract. cited by applicant

Ma et al., "Selective histone deacetylase 6 inhibitors: research advances," *J. Int. Pharm. Res.*, Nov.

2018, 45(11):824-833, with English abstract. cited by applicant
Morfini et al., "Fast Axonal Transport Misregulation and Alzheimer's Disease," *NeuroMolecular Medicine*, 2002, 2:89-99. cited by applicant
Santo et al., "Preclinical activity, pharmacodynamic, and pharmacokinetic properties of a selective HDAC6 inhibitor, ACY-1215, in combination with bortezomib in multiple myeloma," *Blood*, 2012, 119(11):2579-2589. cited by applicant
Selenica et al., "Histone deacetylase 6 inhibition improves memory and reduces total tau levels in a mouse model of tau deposition," *Alzheimer's Research & Therapy*, 2014, 6:12. cited by applicant
Shakespeare et al., "Histone deacetylases as regulators of inflammation and immunity," *Trends in Immunology*, Jul. 2011, 32(7):335-343. cited by applicant
The Practice of Medicinal Chemistry, 2nd. Ed., Wermuth, Ed., 2003, Chapters 9 and 10:131-157. cited by applicant
West et al., "New and emerging HDAC inhibitors for cancer treatment," *The Journal of Clinical Investigation*, Jan. 2014, 124(1):30-39. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Divisional of U.S. application Ser. No. 17/952,546, filed Sep. 26, 2022, now U.S. Pat. No. 11,958,845, which is a Divisional of U.S. application Ser. No. 17/030,504, filed Sep. 24, 2020, now U.S. Pat. No. 11,453,661, which claims priority to JP 2019-177815, filed Sep. 27, 2019.

TECHNICAL FIELD

(1) The present invention relates to a heterocyclic compound having a histone deacetylase (hereinafter sometimes to be referred to as "HDAC") inhibitory activity, preferably a class II HDAC inhibitory activity, more preferably a HDAC6 inhibitory activity, which may be useful for the treatment of central nervous system diseases including neurodegenerative diseases (Alzheimer's disease, progressive supranuclear palsy, etc.) and the like, a medicament comprising the compound, and the like.

BACKGROUND OF THE INVENTION

(2) Nerve axon is known to play an important role in transport of nutritional factor, neurotransmitter, organelle and the like in nerve cell, and axon function disorder, axonal degeneration and intracellular accumulation of axon binding protein tau are observed in various neurodegenerative diseases (Non-Patent Document 1 and Non-Patent Document 2). Diseases characterized by intracellular tau accumulation are collectively called pathologically as tauopathy, and they encompass Alzheimer's disease, progressive supranuclear palsy and the like (Non-Patent Document 3). HDAC6 is an enzyme which plays a role in deacetylation of axon component, tubulin (Non-Patent Document 4), and microtubule containing acetylated tubulin is known to contribute to stability (Non-Patent Document 5). In addition, it is reported that Tubastatin A having a HDAC6 inhibitory activity increases acetylation of tubulin in tauopathy mouse model, and shows therapeutic effectiveness (Non-Patent Document 6). Therefore, the above-mentioned reports suggest that HDAC6 inhibitor has the potential to be a therapeutic drug for Alzheimer's disease and progressive supranuclear palsy via stabilization of axon.

(3) As heterocyclic compounds, for example, the following compounds are known.

(4) (1) Patent Document 1 discloses a compound represented by the following formula:

(5) ##STR00002##

wherein each symbol is as defined in the document, which is a HDAC inhibitor, and is effective for the treatment of autoimmune diseases, inflammatory diseases, metabolic/osteoarticular degenerative diseases, neurodegenerative diseases/central diseases (e.g., schizophrenia, Alzheimer's disease (dementia of Alzheimer type), Parkinson's disease, Huntington's disease, Rubinstein-Taybi syndrome, muscular dystrophy, Rett syndrome, Charcot-Marie-Tooth disease, depression), neoplastic diseases and the like.

(2) Patent Document 2 discloses a compound represented by the following formula:

(6) ##STR00003##

wherein each symbol is as defined in the document, which is a HDAC inhibitor, and is effective for the treatment of autoimmune diseases, inflammatory diseases, metabolic/osteoarticular degenerative diseases, neurodegenerative diseases/central diseases (e.g., schizophrenia, Alzheimer's disease (dementia of Alzheimer type), Parkinson's disease, Huntington's disease, Rubinstein-Taybi syndrome, muscular dystrophy, Rett syndrome, Charcot-Marie-Tooth disease, depression), neoplastic diseases and the like.

(3) Patent Document 3 discloses a compound represented by the following formula:

(7) ##STR00004##

wherein each symbol is as defined in the document, which is a HDAC inhibitor, and is effective for the treatment of autoimmune diseases, inflammatory diseases, metabolic/osteoarticular degenerative diseases, neurodegenerative diseases/central diseases (e.g., schizophrenia, Alzheimer's disease (dementia of Alzheimer type), Parkinson's disease, Huntington's disease, Rubinstein-Taybi syndrome, muscular dystrophy, Rett syndrome, Charcot-Marie-Tooth disease, depression), neoplastic diseases and the like.

(4) Patent Document 4 discloses a compound represented by the following formula:

(8) ##STR00005##

wherein each symbol is as defined in the document, which is a HDAC inhibitor, and is effective for the treatment of autoimmune diseases, inflammatory diseases, metabolic/osteoarticular degenerative diseases, neurodegenerative diseases/central diseases (e.g., schizophrenia, Alzheimer's disease (dementia of Alzheimer type), Parkinson's disease, Huntington's disease, Rubinstein-Taybi syndrome, muscular dystrophy, Rett syndrome, Charcot-Marie-Tooth disease, depression), neoplastic diseases and the like.

(5) Patent Document 5 discloses a compound represented by the following formula:

(9) ##STR00006##

wherein each symbol is as defined in the document, which is a HDAC inhibitor, and is effective for the treatment of autoimmune diseases, inflammatory diseases, metabolic/osteoarticular degenerative diseases, neurodegenerative diseases/central diseases (e.g., schizophrenia, Alzheimer's disease (dementia of Alzheimer type), Parkinson's disease, Huntington's disease, Rubinstein-Taybi syndrome, muscular dystrophy, Rett syndrome, Charcot-Marie-Tooth disease, depression), neoplastic diseases and the like.

(6) Patent Document 6 discloses a compound represented by the following formula:

(10) ##STR00007##

wherein each symbol is as defined in the document, which is a HDAC6 inhibitor, and is effective for the treatment of HDAC6-mediated diseases (e.g., infections, tumor, endocrine/nutritional/metabolic diseases, mental and behavioral disorders, neurological diseases, diseases of the eye and adnexa, cardiovascular diseases, respiratory diseases, digestive diseases, diseases of the skin and subcutaneous tissue, diseases of the musculoskeletal system and connective tissue, or congenital malformations, deformations and chromosomal abnormalities).

(7) Patent Document 7 discloses a compound represented by the following formula:

(11) ##STR00008##

wherein each symbol is as defined in the document, which is a HDAC6 inhibitor, and is effective

for the treatment of HDAC6-mediated diseases (e.g., infections, tumor, endocrine/nutritional/metabolic diseases, mental and behavioral disorders, neurological diseases, diseases of the eye and adnexa, cardiovascular diseases, respiratory diseases, digestive diseases, diseases of the skin and subcutaneous tissue, diseases of the musculoskeletal system and connective tissue, or congenital malformations, deformations and chromosomal abnormalities).

(8) Patent Document 8 discloses a compound represented by the following formula:

(12) ##STR00009##

wherein each symbol is as defined in the document, which is a HDAC6 inhibitor, and is effective for the treatment of HDAC6-mediated diseases (e.g., infections, tumor, endocrine/nutritional/metabolic diseases, mental and behavioral disorders, neurological diseases, diseases of the eye and adnexa, cardiovascular diseases, respiratory diseases, digestive diseases, diseases of the skin and subcutaneous tissue, diseases of the musculoskeletal system and connective tissue, or congenital malformations, deformations and chromosomal abnormalities).

(9) Patent Document 9 discloses a compound represented by the following formula:

(13) ##STR00010##

wherein each symbol is as defined in the document, which is a HDAC6 inhibitor, and is effective for the treatment of HDAC6-mediated diseases (e.g., infections, tumor, endocrine/nutritional/metabolic diseases, mental and behavioral disorders, neurological diseases, diseases of the eye and adnexa, cardiovascular diseases, respiratory diseases, digestive diseases, diseases of the skin and subcutaneous tissue, diseases of the musculoskeletal system and connective tissue, or congenital malformations, deformations and chromosomal abnormalities).

(10) Patent Document 10 discloses a compound represented by the following formula:

(14) ##STR00011##

wherein each symbol is as defined in the document, which is a HDAC inhibitor, and is effective for the treatment of epilepsy, attentional deficit disorder, depression, anxiety, Alzheimer's disease, Parkinson's disease, Huntington's disease and the like.

(11) Patent Document 11 discloses a compound represented by the following formula:

(15) ##STR00012##

wherein each symbol is as defined in the document, which is a HDAC inhibitor, and is effective for the treatment of infectious diseases, tumors, endocrine diseases, nutritional and metabolic diseases and the like.

(12) Patent Document 12 discloses a compound represented by the following formula:

(16) ##STR00013##

wherein each symbol is as defined in the document, which is a HDAC inhibitor, and is effective for the treatment of autoimmune diseases, inflammatory diseases, metabolic/osteoarticular degenerative diseases, neurodegenerative diseases/central diseases (e.g., schizophrenia, Alzheimer's disease (dementia of Alzheimer type), Parkinson's disease, Huntington's disease, Rubinstein-Taybi syndrome, muscular dystrophy, Rett syndrome, Charcot-Marie-Tooth disease, depression), neoplastic diseases and the like.

DOCUMENT LIST

Patent Document

(17) Patent Document 1: WO 2016/031815 Patent Document 2: WO 2017/014321 Patent Document 3: WO 2017/014170 Patent Document 4: WO 2017/033946 Patent Document 5: WO 2019/027054 Patent Document 6: WO 2017/018803 Patent Document 7: WO 2017/018804 Patent Document 8: WO 2017/018805 Patent Document 9: WO 2017/023133 Patent Document 10: WO 2018/165520 Patent Document 11: WO 2020/022794 Patent Document 12: WO 2020/158762

Non-Patent Document

(18) Non-Patent Document 1: Front Cell Neurosci. 9, 343, (2015). Non-Patent Document 2: Neuromolecular Med. 2: 89-99, (2002). Non-Patent Document 3: Cold Spring Harb Perspect Med. 2: a006254 (2012). Non-Patent Document 4: Nature. 417: 455-458, (2002). Non-Patent Document

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(19) The present invention aims to provide a heterocyclic compound having a HDAC inhibitory action, which is useful for the treatment of central nervous system diseases including neurodegenerative diseases (Alzheimer's disease, progressive supranuclear palsy, etc.) and the like, and a medicament comprising the compound.

Means of Solving the Problems

(20) The present inventors have conducted intensive studies to solve the above-mentioned problems, and have found that a compound represented by the following formula (I) has a superior HDAC inhibitory action, and completed the present invention based on these findings.

(21) Accordingly, the present invention provides the following.

(22) [1] A compound represented by the formula (I):

(23) ##STR00014##

wherein R^{sup.1} is an optionally substituted cyclic group, and R^{sup.2} is an optionally substituted cyclic group, or a salt thereof (hereinafter sometimes to be referred to as "compound (I)").

[2] The compound or salt according to the above-mentioned [1], which is represented by the formula (I'):

(24) ##STR00015##

wherein each symbol is as defined in the above-mentioned [1].

[3] The compound or salt according to the above-mentioned [1],

wherein

R^{sup.1} is (1) a C_{sub}.3-10 cycloalkyl group optionally substituted by 1 to 3 halogen atoms, (2) a C_{sub}.6-14 aryl group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a cyano group, and (iii) a C_{sub}.1-6 alkyl group optionally substituted by 1 to 3 halogen atoms, (3) a 5- or 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a C_{sub}.1-6 alkyl group optionally substituted by 1 to 3 halogen atoms, and (iii) a C_{sub}.1-6 alkoxy group, or (4) a 3- to 8-membered monocyclic non-aromatic heterocyclic group, and

R^{sup.2} is (1) a C_{sub}.3-10 cycloalkyl group optionally substituted by 1 to 3 halogen atoms, (2) a C_{sub}.6-14 aryl group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, and (ii) a C_{sub}.1-6 alkyl group optionally substituted by 1 to 3 halogen atoms, (3) a 5- or 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a C_{sub}.1-6 alkyl group optionally substituted by 1 to 3 halogen atoms, and (iii) a C_{sub}.1-6 alkoxy group, or (4) a 3- to 8-membered monocyclic non-aromatic heterocyclic group.

[4] The compound or salt according to the above-mentioned [1],

wherein

R^{sup.1} is (1) a C_{sub}.3-10 cycloalkyl group optionally substituted by 1 to 3 halogen atoms, or (2) a 5- or 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a C_{sub}.1-6 alkyl group optionally substituted by 1 to 3 halogen atoms, and (iii) a C_{sub}.1-6 alkoxy group, and

R^{sup.2} is (1) a C_{sub}.6-14 aryl group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, and (ii) a C_{sub}.1-6 alkyl group optionally substituted by 1 to 3 halogen atoms, or (2) a 5- or 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a C_{sub}.1-6 alkyl group optionally substituted by 1 to 3 halogen atoms, and (iii) a C_{sub}.1-6 alkoxy group.

[5] The compound or salt according to the above-mentioned [1],

wherein

R.sup.1 is (1) a C.sub.3-10 cycloalkyl group, or (2) a 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 halogen atoms, and
R.sup.2 is (1) a C.sub.6-14 aryl group optionally substituted by 1 to 3 halogen atoms, or (2) a 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 halogen atoms.
[6] 6-[5-(Difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one or a salt thereof.
[7] 6-[5-(Difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one or a salt thereof.
[8] 2-[(1R,2R)-1,2-Bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one or a salt thereof.
[9] A medicament comprising the compound or salt according to the above-mentioned [1].
[10] The medicament according to the above-mentioned [9], which is a histone deacetylase 6 inhibitor.
[11] The medicament according to the above-mentioned [9], which is an agent for the prophylaxis or treatment of Alzheimer's disease or progressive supranuclear palsy.
[12] The compound or salt according to the above-mentioned [1] for use in the prophylaxis or treatment of Alzheimer's disease or progressive supranuclear palsy.
[13] A method of inhibiting histone deacetylase 6 in a mammal, which comprises administering an effective amount of the compound or salt according to the above-mentioned [1] to the mammal.
[14] A method for the prophylaxis or treatment of Alzheimer's disease or progressive supranuclear palsy in a mammal, which comprises administering an effective amount of the compound or salt according to the above-mentioned [1] to the mammal.
[15] Use of the compound or salt according to the above-mentioned [1] for the production of an agent for the prophylaxis or treatment of Alzheimer's disease or progressive supranuclear palsy.
Effect of the Invention
(25) Compound (I) has a HDAC6 inhibitory activity, and may be useful for the treatment of central nervous system diseases including neurodegenerative diseases (Alzheimer's disease, progressive supranuclear palsy, etc.) and the like.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a graph showing increase in acetylated tubulin in mouse brain by the compound of Example 1. The vertical axis is relative tubulin acetylation level, and the horizontal axis is the dose (mg/kg).
(2) FIG. 2 is a graph showing increase in acetylated tubulin in mouse brain by the compound of Example 3. The vertical axis is relative tubulin acetylation level, and the horizontal axis is the dose (mg/kg).
(3) FIG. 3 is a graph showing increase in acetylated tubulin in mouse brain by the compound of Example 5. The vertical axis is relative tubulin acetylation level, and the horizontal axis is the dose (mg/kg).
(4) FIG. 4 is a graph showing cognitive improvement action in mouse by the compound of Example 1. The vertical axis is relative exploration time of the novelty discrimination (%), and the horizontal axis is the dose (mg/kg).
(5) FIG. 5 is a graph showing cognitive improvement action in mouse by the compound of Example 3. The vertical axis is the novelty discrimination relative exploration time of (%), and the horizontal axis is the dose (mg/kg).

DETAILED DESCRIPTION OF THE INVENTION

- (6) The present invention is explained in detail in the following.

(7) The definition of each symbol used in the formula (I) is explained.

(8) R_{sup}.1 and R_{sup}.2 are the same or different and each is an optionally substituted cyclic group. Examples of the “cyclic group” of the “optionally substituted cyclic group” include a hydrocarbon ring group and a heterocyclic group.

(9) Examples of the “hydrocarbon ring group” include a C_{sub}.3-10 cycloalkyl group (e.g., cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, cyclooctyl, bicyclo[2.2.1]heptyl, bicyclo[2.2.2]octyl, bicyclo[3.2.1]octyl, adamantyl, etc.), a C_{sub}.3-10 cycloalkenyl group (e.g., cyclopropenyl, cyclobutenyl, cyclopentenyl, cyclohexenyl, cycloheptenyl, cyclooctenyl), a C_{sub}.6-14 aryl group (e.g., phenyl, 1-naphthyl, 2-naphthyl, 1-anthryl, 2-anthryl, 9-anthryl) and the like.

(10) Examples of the “heterocyclic group” include (i) an aromatic heterocyclic group, (ii) a non-aromatic heterocyclic group and (iii) a 7- to 10-membered bridged heterocyclic group, each containing, as a ring-constituting atom besides carbon atom, 1 to 4 heteroatoms selected from a nitrogen atom, a sulfur atom and an oxygen atom.

(11) Examples of the “aromatic heterocyclic group” include a 5- to 14-membered (preferably 5- to 10-membered) aromatic heterocyclic group containing, as a ring-constituting atom besides carbon atom, 1 to 4 heteroatoms selected from a nitrogen atom, a sulfur atom and an oxygen atom.

(12) Preferable examples of the “aromatic heterocyclic group” include 5- or 6-membered monocyclic aromatic heterocyclic groups such as thienyl, furyl, pyrrolyl, imidazolyl, pyrazolyl, thiazolyl, isothiazolyl, oxazolyl, isoxazolyl, pyridyl, pyrazinyl, pyrimidinyl, pyridazinyl, 1,2,4-oxadiazolyl, 1,3,4-oxadiazolyl, 1,2,4-thiadiazolyl, 1,3,4-thiadiazolyl, triazolyl, tetrazolyl, triazinyl and the like; and 8- to 14-membered fused polycyclic (preferably bi or tricyclic) aromatic heterocyclic groups such as benzothiophenyl, benzofuranyl, benzimidazolyl, benzoxazolyl, benzisoxazolyl, benzothiazolyl, benzisothiazolyl, benzotriazolyl, imidazopyridinyl, thienopyridinyl, furopyridinyl, pyrrolopyridinyl, pyrazolopyridinyl, oxazolopyridinyl, thiazolopyridinyl, imidazopyrazinyl, imidazopyrimidinyl, thienopyrimidinyl, furopyrimidinyl, pyrrolopyrimidinyl, pyrazolopyrimidinyl, oxazolopyrimidinyl, thiazolopyrimidinyl, pyrazolotriazinyl, naphtho[2,3-b]thienyl, phenoxathiinyl, indolyl, isoindolyl, 1H-indazolyl, purinyl, isoquinolyl, quinolyl, phthalazinyl, naphthyridinyl, quinoxalinyl, quinazolinyl, cinnolinyl, carbazolyl, β -carbolinyl, phenanthridinyl, acridinyl, phenazinyl, phenothiazinyl, phenoxazinyl and the like.

(13) Examples of the “non-aromatic heterocyclic group” include a 3- to 14-membered (preferably 4- to 10-membered) non-aromatic heterocyclic group containing, as a ring-constituting atom besides carbon atom, 1 to 4 heteroatoms selected from a nitrogen atom, a sulfur atom and an oxygen atom.

(14) Preferable examples of the “non-aromatic heterocyclic group” include 3- to 8-membered monocyclic non-aromatic heterocyclic groups such as aziridinyl, oxiranyl, thiiranyl, azetidiny, oxetanyl, thietanyl, tetrahydrothienyl, tetrahydrofuranyl, pyrrolinyl, pyrrolidinyl, imidazolinyl, imidazolidinyl, oxazolinyl, oxazolidinyl, pyrazolinyl, pyrazolidinyl, thiazolinyl, thiazolidinyl, tetrahydroisothiazolyl, tetrahydrooxazolyl, tetrahydroisooxazolyl, piperidinyl, piperazinyl, tetrahydropyridinyl, dihydropyridinyl, dihydrothiopyranyl, tetrahydropyrimidinyl, tetrahydropyridazinyl, dihydropyranyl, tetrahydropyranyl, tetrahydrothiopyranyl, morpholinyl, thiomorpholinyl, azepanyl, diazepanyl, azepinyl, oxepanyl, azocanyl, diazocanyl and the like; and 9- to 14-membered fused polycyclic (preferably bi or tricyclic) non-aromatic heterocyclic groups such as dihydrobenzofuranyl, dihydrobenzimidazolyl, dihydrobenzoxazolyl, dihydrobenzothiazolyl, dihydrobenzisothiazolyl, dihydronaphtho[2,3-b]thienyl, tetrahydroisoquinolyl, tetrahydroquinolyl, 4H-quinoliziny, indolinyl, isoindolinyl, tetrahydrothieno[2,3-c]pyridinyl, tetrahydrobenzazepinyl, tetrahydroquinoxalinyl, tetrahydrophenanthridinyl, hexahydrophenothiazinyl, hexahydrophenoxazinyl, tetrahydrophthalazinyl, tetrahydronaphthyridinyl, tetrahydroquinazolinyl, tetrahydrocinnolinyl,

tetrahydrothiazolyl, tetrahydro- β -carbolinyl, tetrahydroacrydiny, tetrahydrophenazinyl, tetrahydrothioxanthenyl, octahydroisoquinolyl and the like.

(15) Preferable examples of the “7- to 10-membered bridged heterocyclic group” include quinuclidinyl and 7-azabicyclo[2.2.1]heptanyl.

(16) Examples of the “optionally substituted cyclic group” represented by R^{sup.1} or R^{sup.2} include a cyclic group optionally having substituent(s) selected from the following Substituent group A.

(17) The number of the substituents in the “optionally substituted cyclic group” is, for example, 1 to 3. When the number of the substituents is two or more, the respective substituents may be the same or different.

(18) [Substituent Group A]

(19) (1) a halogen atom (e.g., fluorine, chlorine, bromine, iodine), (2) a nitro group, (3) a cyano group, (4) an oxo group, (5) a hydroxy group, (6) an optionally halogenated C_{sub.1-6} alkoxy group (e.g., methoxy, difluoromethoxy, trifluoromethoxy, ethoxy, 2,2,2-trifluoroethoxy, propoxy, isopropoxy, butoxy, 4,4,4-trifluorobutoxy, isobutoxy, sec-butoxy, pentyloxy, hexyloxy), (7) a C_{sub.6-14} aryloxy group (e.g., phenoxy, naphthoxy), (8) a C_{sub.7-16} aralkyloxy group (e.g., benzyloxy), (9) a 5- to 14-membered aromatic heterocyclyloxy group (e.g., pyridyloxy), (10) a 3- to 14-membered non-aromatic heterocyclyloxy group (e.g., morpholinyl, piperidinyl), (11) a C_{sub.1-6} alkyl-carbonyloxy group (e.g., acetoxyl, propanoyloxy), (12) a C_{sub.6-14} aryl-carbonyloxy group (e.g., benzoyloxy, 1-naphthoyloxy, 2-naphthoyloxy), (13) a C_{sub.1-6} alkoxy-carbonyloxy group (e.g., methoxycarbonyloxy, ethoxycarbonyloxy, propoxycarbonyloxy, butoxycarbonyloxy), (14) a mono- or di-C_{sub.1-6} alkyl-carbamoyloxy group (e.g., methylcarbamoyloxy, ethylcarbamoyloxy, dimethylcarbamoyloxy, diethylcarbamoyloxy), (15) a C_{sub.6-14} aryl-carbamoyloxy group (e.g., phenylcarbamoyloxy, naphthylcarbamoyloxy), (16) a 5- to 14-membered aromatic heterocyclylcarbonyloxy group (e.g., nicotinoyloxy), (17) a 3- to 14-membered non-aromatic heterocyclylcarbonyloxy group (e.g., morpholinylcarbonyloxy, piperidinylcarbonyloxy), (18) an optionally halogenated C_{sub.1-6} alkylsulfonyloxy group (e.g., methylsulfonyloxy, trifluoromethylsulfonyloxy), (19) a C_{sub.6-14} arylsulfonyloxy group optionally substituted by C_{sub.1-6} alkyl group(s) (e.g., phenylsulfonyloxy, toluenesulfonyloxy), (20) an optionally halogenated C_{sub.1-6} alkylthio group (e.g., methylthio, difluoromethylthio, trifluoromethylthio, ethylthio, propylthio, isopropylthio, butylthio, 4,4,4-trifluorobutylthio, pentylthio, hexylthio), (21) a 5- to 14-membered aromatic heterocyclic group (e.g., thienyl, furyl, pyrrolyl, imidazolyl, pyrazolyl, thiazolyl, isothiazolyl, oxazolyl, isoxazolyl, pyridyl, pyrazinyl, pyrimidinyl, pyridazinyl, 1,2,4-oxadiazolyl, 1,3,4-oxadiazolyl, 1,2,4-thiadiazolyl, 1,3,4-thiadiazolyl, triazolyl, tetrazolyl, triazinyl, benzothiophenyl, benzofuranyl, benzimidazolyl, benzoxazolyl, benzisoxazolyl, benzothiazolyl, benzisothiazolyl, benzotriazolyl, imidazopyridinyl, thienopyridinyl, furopyridinyl, pyrrolopyridinyl, pyrazolopyridinyl, oxazolopyridinyl, thiazolopyridinyl, imidazopyrazinyl, imidazopyrimidinyl, thienopyrimidinyl, furopyrimidinyl, pyrrolopyrimidinyl, pyrazolopyrimidinyl, oxazolopyrimidinyl, thiazolopyrimidinyl, pyrazolotriazinyl, naphtho[2,3-b]thienyl, phenoxathiinyl, indolyl, isoindolyl, 1H-indazolyl, purinyl, isoquinolyl, quinolyl, phthalazinyl, naphthyridinyl, quinoxalinyl, quinazolinyl, cinnolinyl, carbazolyl, β -carbolinyl, phenanthridinyl, acridinyl, phenazinyl, phenothiazinyl, phenoxazinyl), (22) a 3- to 14-membered non-aromatic heterocyclic group (e.g., aziridinyl, oxiranyl, thiranyl, azetidyl, oxetanyl, thietanyl, tetrahydrothienyl, tetrahydrofuranyl, pyrrolinyl, pyrrolidinyl, imidazolinyl, imidazolidinyl, oxazolinyl, oxazolidinyl, pyrazolinyl, pyrazolidinyl, thiazolinyl, thiazolidinyl, tetrahydroisothiazolyl, tetrahydrooxazolyl, tetrahydroisoxazolyl, piperidinyl, piperazinyl, tetrahydropyridinyl, dihydropyridinyl, dihydrothiopyranyl, tetrahydropyrimidinyl, tetrahydropyridazinyl, dihydropyranyl, tetrahydropyranyl, tetrahydrothiopyranyl, morpholinyl, thiomorpholinyl, azepanyl, diazepanyl, azepinyl, oxepanyl, azocanyl, diazocanyl, dihydrobenzofuranyl, dihydrobenzimidazolyl, dihydrobenzoxazolyl, dihydrobenzothiazolyl,

dihydrobenzothiazolyl, dihydronaphtho[2,3-b]thienyl, tetrahydroisoquinolyl, tetrahydroquinolyl, 4H-quinoliziny, indoliny, isoindoliny, tetrahydrothieno[2,3-c]pyridiny, tetrahydrobenzazepiny, tetrahydroquinoxaliny, tetrahydrophenanthridiny, hexahydrophenothiaziny, hexahydrophenoxaziny, tetrahydrophthalaziny, tetrahydronaphthyridiny, tetrahydroquinazoliny, tetrahydrocinnoliny, tetrahydrocarbazolyl, tetrahydro- β -carboliny, tetrahydroacridiny, tetrahydrophenaziny, tetrahydrothioxanthenyl, octahydroisoquinolyl), (23) a formyl group, (24) a carboxy group, (25) an optionally halogenated C.sub.1-6 alkyl-carbonyl group (e.g., acetyl, chloroacetyl, trifluoroacetyl, trichloroacetyl, propanoyl, butanoyl, pentanoyl, hexanoyl), (26) a C.sub.6-14 aryl-carbonyl group (e.g., benzoyl, 1-naphthoyl, 2-naphthoyl), (27) a 5- to 14-membered aromatic heterocyclylcarbonyl group (e.g., nicotinoyl, isonicotinoyl, thenoyl, furoyl), (28) a 3- to 14-membered non-aromatic heterocyclylcarbonyl group (e.g., morpholinylcarbonyl, piperidinylcarbonyl, pyrrolidinylcarbonyl), (29) a C.sub.1-6 alkoxy-carbonyl group (e.g., methoxycarbonyl, ethoxycarbonyl, propoxycarbonyl, isopropoxycarbonyl, butoxycarbonyl, isobutoxycarbonyl, sec-butoxycarbonyl, tert-butoxycarbonyl, pentyloxycarbonyl, hexyloxycarbonyl), (30) a C.sub.6-14 aryloxy-carbonyl group (e.g., phenyloxycarbonyl, 1-naphthyloxycarbonyl, 2-naphthyloxycarbonyl), (31) a C.sub.7-16 aralkyloxy-carbonyl group (e.g., benzyloxycarbonyl, phenethyloxycarbonyl), (32) a carbamoyl group, (33) a thiocarbamoyl group, (34) a mono- or di-C.sub.1-6 alkyl-carbamoyl group (e.g., methylcarbamoyl, ethylcarbamoyl, dimethylcarbamoyl, diethylcarbamoyl, N-ethyl-N-methylcarbamoyl), (35) a C.sub.6-14 aryl-carbamoyl group (e.g., phenylcarbamoyl), (36) a 5- to 14-membered aromatic heterocyclylcarbamoyl group (e.g., pyridylcarbamoyl, thienylcarbamoyl), (37) a 3- to 14-membered non-aromatic heterocyclylcarbamoyl group (e.g., morpholinylcarbamoyl, piperidinylcarbamoyl), (38) an optionally halogenated C.sub.1-6 alkylsulfonyl group (e.g., methylsulfonyl, difluoromethylsulfonyl, trifluoromethylsulfonyl, ethylsulfonyl, propylsulfonyl, isopropylsulfonyl, butylsulfonyl, 4,4,4-trifluorobutylsulfonyl, pentylsulfonyl, hexylsulfonyl), (39) a C.sub.6-14 arylsulfonyl group (e.g., phenylsulfonyl, 1-naphthylsulfonyl, 2-naphthylsulfonyl), (40) a 5- to 14-membered aromatic heterocyclylsulfonyl group (e.g., pyridylsulfonyl, thienylsulfonyl), (41) an optionally halogenated C.sub.1-6 alkylsulfinyl group (e.g., methylsulfinyl, ethylsulfinyl), (42) a C.sub.6-14 aryl sulfinyl group (e.g., phenylsulfinyl, 1-naphthyl sulfinyl, 2-naphthyl sulfinyl), (43) a 5- to 14-membered aromatic heterocyclylsulfinyl group (e.g., pyridyl sulfinyl, thienylsulfinyl), (44) an amino group, (45) a mono- or di-C.sub.1-6 alkylamino group (e.g., methylamino, ethylamino, propylamino, isopropylamino, butylamino, dimethylamino, diethylamino, dipropylamino, dibutylamino, N-ethyl-N-methylamino), (46) a mono- or di-C.sub.6-14 arylamino group (e.g., phenylamino), (47) a 5- to 14-membered aromatic heterocyclylamino group (e.g., pyridylamino), (48) a C.sub.7-16 aralkylamino group (e.g., benzylamino), (49) a formylamino group, (50) a C.sub.1-6 alkyl-carbonylamino group (e.g., acetilamino, propanoylamino, butanoylamino), (51) a (C.sub.1-6 alkyl) (C.sub.1-6 alkyl-carbonyl) amino group (e.g., N-acetyl-N-methylamino), (52) a C.sub.6-14 aryl-carbonylamino group (e.g., phenylcarbonylamino, naphthylcarbonylamino), (53) a C.sub.1-6 alkoxy-carbonylamino group (e.g., methoxycarbonylamino, ethoxycarbonylamino, propoxycarbonylamino, butoxycarbonylamino, tert-butoxycarbonylamino), (54) a C.sub.7-16 aralkyloxy-carbonylamino group (e.g., benzyloxycarbonylamino), (55) a C.sub.1-6 alkylsulfonylamino group (e.g., methylsulfonylamino, ethylsulfonylamino), (56) a C.sub.6-14 arylsulfonylamino group optionally substituted by C.sub.1-6 alkyl group(s) (e.g., phenylsulfonylamino, toluenesulfonylamino), (57) an optionally halogenated C.sub.1-6 alkyl group (e.g., methyl, chloromethyl, difluoromethyl, trichloromethyl, trifluoromethyl, ethyl, 2-bromoethyl, 2,2,2-trifluoroethyl, tetrafluoroethyl, pentafluoroethyl, propyl, 2,2-difluoropropyl, 3,3,3-trifluoropropyl, isopropyl, butyl, 4,4,4-trifluorobutyl, isobutyl, sec-butyl, tert-butyl, pentyl, isopentyl, neo-pentyl, 5,5,5-trifluoropentyl, hexyl, 6,6,6-trifluorohexyl), (58) a C.sub.2-6 alkenyl group (e.g., ethenyl, 1-propenyl, 2-propenyl, 2-methyl-1-propenyl, 1-butenyl, 2-butenyl, 3-butenyl, 3-methyl-2-butenyl, 1-pentenyl, 2-pentenyl,

3-pentenyl, 4-pentenyl, 4-methyl-3-pentenyl, 1-hexenyl, 3-hexenyl, 5-hexenyl), (59) a C.sub.2-6 alkynyl group (e.g., ethynyl, 1-propynyl, 2-propynyl, 1-butylnyl, 2-butylnyl, 3-butylnyl, 1-pentynyl, 2-pentynyl, 3-pentynyl, 4-pentynyl, 1-hexynyl, 2-hexynyl, 3-hexynyl, 4-hexynyl, 5-hexynyl, 4-methyl-2-pentynyl), (60) a C.sub.3-10 cycloalkyl group (e.g., cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, cyclooctyl, bicyclo[2.2.1]heptyl, bicyclo[2.2.2]octyl, bicyclo[3.2.1]octyl, adamantyl), (61) a C.sub.3-10 cycloalkenyl group (e.g., cyclopropenyl, cyclobutenyl, cyclopentenyl, cyclohexenyl, cycloheptenyl, cyclooctenyl), and (62) a C.sub.6-14 aryl group (e.g., phenyl, 1-naphthyl, 2-naphthyl, 1-anthryl, 2-anthryl, 9-anthryl).

(20) R.sup.1 is preferably (1) a C.sub.3-10 cycloalkyl group (e.g., cyclopropyl, cyclohexyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), (2) a C.sub.6-14 aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom), (ii) a cyano group, and (iii) a C.sub.1-6 alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), (3) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyrazolyl, pyridyl, pyrimidinyl, pyrazinyl, pyridazinyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom), (ii) a C.sub.1-6 alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and (iii) a C.sub.1-6 alkoxy group (e.g., ethoxy), or (4) a 3- to 8-membered monocyclic non-aromatic heterocyclic group (e.g., tetrahydropyranyl).

(21) R.sup.1 is more preferably (1) a C.sub.3-10 cycloalkyl group (e.g., cyclopropyl, cyclohexyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), or (2) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyrazolyl, pyridyl, pyrimidinyl, pyrazinyl, pyridazinyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom), (ii) a C.sub.1-6 alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and (iii) a C.sub.1-6 alkoxy group (e.g., ethoxy).

(22) R.sup.1 is particularly preferably (1) a C.sub.3-10 cycloalkyl group (e.g., cyclopropyl, cyclohexyl), or (2) a 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl, pyrazinyl, pyridazinyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom).

(23) R.sup.2 is preferably (1) a C.sub.3-10 cycloalkyl group (e.g., cyclopropyl, cyclohexyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), (2) a C.sub.6-14 aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), and (ii) a C.sub.1-6 alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), (3) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), (ii) a C.sub.1-6 alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and (iii) a C.sub.1-6 alkoxy group (e.g., ethoxy), or (4) a 3- to 8-membered monocyclic non-aromatic heterocyclic group (e.g., tetrahydropyranyl).

(24) R.sup.2 is more preferably (1) a C.sub.6-14 aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), and (ii) a C.sub.1-6 alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), or (2) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), (ii) a C.sub.1-6 alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and (iii) a C.sub.1-6 alkoxy group (e.g., ethoxy).

(25) R.sup.2 is particularly preferably (1) a C.sub.6-14 aryl group (e.g., phenyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom, a chlorine atom), or (2) a 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom, a chlorine atom).

(26) Preferred compound is compound (I) wherein

(27) R^{sup.1} is

(28) (1) a C_{sub.3-10} cycloalkyl group (e.g., cyclopropyl, cyclohexyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), (2) a C_{sub.6-14} aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom), (ii) a cyano group, and (iii) a C_{sub.1-6} alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), (3) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyrazolyl, pyridyl, pyrimidinyl, pyrazinyl, pyridazinyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom), (ii) a C_{sub.1-6} alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and (iii) a C_{sub.1-6} alkoxy group (e.g., ethoxy), or (4) a 3- to 8-membered monocyclic non-aromatic heterocyclic group (e.g., tetrahydropyranyl), and R^{sup.2} is (1) a C_{sub.3-10} cycloalkyl group (e.g., cyclopropyl, cyclohexyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), (2) a C_{sub.6-14} aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), and (ii) a C_{sub.1-6} alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), (3) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), (ii) a C_{sub.1-6} alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and (iii) a C_{sub.1-6} alkoxy group (e.g., ethoxy), or (4) a 3- to 8-membered monocyclic non-aromatic heterocyclic group (e.g., tetrahydropyranyl).

(29) More preferred compound is compound (I) wherein

(30) R^{sup.1} is

(31) (1) a C_{sub.3-10} cycloalkyl group (e.g., cyclopropyl, cyclohexyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), or (2) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyrazolyl, pyridyl, pyrimidinyl, pyrazinyl, pyridazinyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom), (ii) a C_{sub.1-6} alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and (iii) a C_{sub.1-6} alkoxy group (e.g., ethoxy), and

R^{sup.2} is (1) a C_{sub.6-14} aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), and (ii) a C_{sub.1-6} alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), or (2) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), (ii) a C_{sub.1-6} alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and (iii) a C_{sub.1-6} alkoxy group (e.g., ethoxy).

(32) Particularly preferred compound is compound (I) wherein

(33) R^{sup.1} is

(34) (1) a C_{sub.3-10} cycloalkyl group (e.g., cyclopropyl, cyclohexyl), or (2) a 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl, pyrazinyl, pyridazinyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and

R^{sup.2} is (1) a C_{sub.6-14} aryl group (e.g., phenyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom, a chlorine atom), or (2) a 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom, a chlorine atom).

(35) The configuration of compound (I) is preferably represented by the formula (I'):

(36) ##STR00016##

wherein each symbol is as defined above,

(hereinafter sometimes to be referred to as compound (I')).

(37) As embodiments of compound (I), the following compounds (compound (I'-1) and compound (I'-2)) are exemplified.

(38) Compound (I'-1)

(39) Compound (I') wherein R.sup.1 is an optionally substituted hydrocarbon ring group, and R.sup.2 is an optionally substituted hydrocarbon ring group.

Compound (I'-1a)

(40) Compound (I') wherein

(41) R.sup.1 is

(42) (1) a C.sub.3-10 cycloalkyl group (e.g., cyclopropyl, cyclohexyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), or (2) a C.sub.6-14 aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom), (ii) a cyano group, and (iii) a C.sub.1-6 alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and

R.sup.2 is (1) a C.sub.3-10 cycloalkyl group (e.g., cyclopropyl, cyclohexyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), or (2) a C.sub.6-14 aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), and (ii) a C.sub.1-6 alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom).

Compound (I'-1b)

(43) Compound (I') wherein R.sup.1 is a C.sub.3-10 cycloalkyl group (e.g., cyclopropyl, cyclohexyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and R.sup.2 is a C.sub.6-14 aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), and (ii) a C.sub.1-6 alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom).

Compound (I'-1c)

(44) Compound (I') wherein R.sup.1 is a C.sub.3-10 cycloalkyl group (e.g., cyclopropyl, cyclohexyl), and R.sup.2 is a C.sub.6-14 aryl group (e.g., phenyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom, a chlorine atom).

Compound (I'-2)

(45) Compound (I') wherein R.sup.1 is an optionally substituted heterocyclic group, and R.sup.2 is an optionally substituted heterocyclic group.

Compound (I'-2a)

(46) Compound (I') wherein

(47) R.sup.1 is

(48) (1) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyrazolyl, pyridyl, pyrimidinyl, pyrazinyl, pyridazinyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom), (ii) a C.sub.1-6 alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and (iii) a C.sub.1-6 alkoxy group (e.g., ethoxy), or (2) a 3- to 8-membered monocyclic non-aromatic heterocyclic group (e.g., tetrahydropyranyl), and

R.sup.2 is (1) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), (ii) a C.sub.1-6 alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and (iii) a C.sub.1-6 alkoxy group (e.g., ethoxy), or (2) a 3- to 8-membered monocyclic non-aromatic heterocyclic group (e.g., tetrahydropyranyl).

Compound (I'-2b)

(49) Compound (I') wherein R.sup.1 is a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyrazolyl, pyridyl, pyrimidinyl, pyrazinyl, pyridazinyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom), (ii) a C.sub.1-6 alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and (iii) a C.sub.1-6 alkoxy group (e.g., ethoxy), and R.sup.2 is a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl) optionally substituted by 1 to 3 substituents selected

from (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), (ii) a C.sub.1-6 alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and (iii) a C.sub.1-6 alkoxy group (e.g., ethoxy).

Compound (I'-2c)

(50) Compound (I') wherein R^{sup.1} is a 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl, pyrazinyl, pyridazinyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and R^{sup.2} is a 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom, a chlorine atom).

(51) Specific examples of compound (I) include the compounds of Examples 1 to 150. Among them, compound (I) is preferably 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Example 1) or a salt thereof, 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Example 3) or a salt thereof, 2-[(1R,2R)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Example 5) or a salt thereof, 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Example 7) or a salt thereof, or 2-[(1R,2S)-1-cyclopropyl-2-(4-fluorophenyl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Example 9) or a salt thereof, more preferably 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Example 1) or a salt thereof, 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Example 3) or a salt thereof, or 2-[(1R,2R)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Example 5) or a salt thereof.

(52) When compound (I) is in a form of a salt, examples thereof include metal salts, an ammonium salt, salts with organic base, salts with inorganic acid, salts with organic acid, salts with basic or acidic amino acid, and the like. Preferable examples of the metal salt include alkali metal salts such as sodium salt, potassium salt and the like; alkaline earth metal salts such as calcium salt, magnesium salt, barium salt and the like; an aluminum salt, and the like. Preferable examples of the salt with organic base include salts with trimethylamine, triethylamine, pyridine, picoline, 2,6-lutidine, ethanolamine, diethanolamine, triethanolamine, cyclohexylamine, dicyclohexylamine, N,N'-dibenzylethylenediamine and the like. Preferable examples of the salt with inorganic acid include salts with hydrochloric acid, hydrobromic acid, nitric acid, sulfuric acid, phosphoric acid and the like. Preferable examples of the salt with organic acid include salts with formic acid, acetic acid, trifluoroacetic acid, phthalic acid, fumaric acid, oxalic acid, tartaric acid, maleic acid, citric acid, succinic acid, malic acid, methanesulfonic acid, benzenesulfonic acid, p-toluenesulfonic acid and the like. Preferable examples of the salt with basic amino acid include salts with arginine, lysine, ornithine and the like. Preferable examples of the salt with acidic amino acid include salts with aspartic acid, glutamic acid and the like.

(53) Among them, a pharmaceutically acceptable salt is preferable. For example, when a compound has an acidic functional group, examples thereof include inorganic salts such as alkali metal salts (e.g., sodium salt, potassium salt etc.), alkaline earth metal salts (e.g., calcium salt, magnesium salt etc.) and the like, ammonium salt etc., and when a compound has a basic functional group, examples thereof include salts with inorganic acid such as hydrochloric acid, hydrobromic acid, nitric acid, sulfuric acid, phosphoric acid and the like, and salts with organic acid such as acetic acid, phthalic acid, fumaric acid, oxalic acid, tartaric acid, maleic acid, citric acid, succinic acid, methanesulfonic acid, benzenesulfonic acid, p-toluenesulfonic acid and the like.

(54) [Production Method]

(55) The production method of compound (I) is explained below.

(56) The raw material compound and reagent used and the compound obtained in each step in the following production method may be each in a form of a salt, and examples of such salt include those similar to the salts of compound (I) and the like.

(57) When the compound obtained in each step is a free form, it can be converted to the objective salt according to a method known per se. When the compound obtained in each step is a salt, it can be converted to the objective free form or the other salt according to a method known per se.

(58) The compound obtained in each step can be used directly as the reaction mixture or as a crude product for the next reaction. Alternatively, the compound obtained in each step can be isolated and purified from a reaction mixture according to a method known per se, for example, a separation means such as concentration, crystallization, recrystallization, distillation, solvent extraction, fractional distillation, chromatography and the like.

(59) When the raw material compound and reagent used in each step are commercially available, the commercially available product can also be used directly.

(60) In the reaction in each step, while the reaction time varies depending on the kind of the reagent and solvent to be used, it is generally 1 min-48 hr, preferably 10 min-8 hr, unless otherwise specified.

(61) In the reaction in each step, while the reaction temperature varies depending on the kind of the reagent and solvent to be used, it is generally -78°C. - 300°C. , preferably -78°C. - 150°C. , unless otherwise specified.

(62) In the reaction in each step, while the pressure varies depending on the kind of the reagent and solvent to be used, it is generally 1 atm-20 atm, preferably 1 atm-3 atm, unless otherwise specified.

(63) Microwave synthesizer such as Initiator manufactured by Biotage and the like may be used for the reaction in each step. While the reaction temperature varies depending on the kind of the reagent and solvent to be used, it is generally room temperature— 300°C. , preferably 50°C. - 250°C. , unless otherwise specified. While the reaction time varies depending on the kind of the reagent and solvent to be used, it is generally 1 min-48 hr, preferably 1 min-8 hr, unless otherwise specified.

(64) In the reaction in each step, the reagent is used in an amount of 0.5 equivalents-20 equivalents, preferably 0.8 equivalents-5 equivalents, relative to the substrate, unless otherwise specified. When the reagent is used as a catalyst, the reagent is used in an amount of 0.001 equivalent-1 equivalent, preferably 0.01 equivalent-0.2 equivalent, relative to the substrate. When the reagent is used as a reaction solvent, the reagent is used in a solvent amount.

(65) Unless otherwise specified, the reaction in each step is carried out without solvent, or by dissolving or suspending the raw material compound in a suitable solvent. Examples of the solvent include those described in Examples and the following solvents. alcohols: methanol, ethanol, tert-butyl alcohol, 2-methoxyethanol and the like; ethers: diethyl ether, diphenyl ether, tetrahydrofuran, 1,2-dimethoxyethane and the like; aromatic hydrocarbons: chlorobenzene, toluene, xylene and the like; saturated hydrocarbons: cyclohexane, hexane and the like; amides: N,N-dimethylformamide, N-methylpyrrolidone and the like; halogenated hydrocarbons: dichloromethane, carbon tetrachloride and the like; nitriles: acetonitrile and the like; sulfoxides: dimethyl sulfoxide and the like; aromatic organic bases: pyridine and the like; anhydrides: acetic anhydride and the like; organic acids: formic acid, acetic acid, trifluoroacetic acid and the like; inorganic acids: hydrochloric acid, sulfuric acid and the like; esters: ethyl acetate and the like; ketones: acetone, methyl ethyl ketone and the like; water.

(66) The above-mentioned solvent can be used in a mixture of two or more kinds thereof in an appropriate ratio.

(67) When a base is used for the reaction in each step, examples thereof include those described in Examples and the following bases. inorganic bases: sodium hydroxide, magnesium hydroxide, sodium carbonate, calcium carbonate, sodium hydrogencarbonate, cesium carbonate, potassium acetate and the like; organic bases: triethylamine, diethylamine, N,N-diisopropylethylamine

(DIPEA), pyridine, 4-dimethylaminopyridine, N,N-dimethylaniline, 1,4-diazabicyclo[2.2.2]octane, 1,8-diazabicyclo[5.4.0]-7-undecene, imidazole, piperidine and the like; metal alkoxides: sodium ethoxide, potassium tert-butoxide and the like; alkali metal hydrides: sodium hydride and the like; metal amides: sodium amide, lithium diisopropylamide, lithium hexamethyldisilazide and the like; organic lithiums: n-butyllithium and the like.

(68) When an acid or an acid catalyst is used for the reaction in each step, examples thereof include those described in Examples and the following acids and acid catalysts. inorganic acids: hydrochloric acid, sulfuric acid, nitric acid, hydrobromic acid, phosphoric acid and the like; organic acids: acetic acid, trifluoroacetic acid, citric acid, p-toluenesulfonic acid, 10-camphorsulfonic acid and the like; Lewis acid: boron trifluoride diethyl ether complex, zinc iodide, anhydrous aluminium chloride, anhydrous zinc chloride, anhydrous iron chloride and the like.

(69) Unless otherwise specified, the reaction in each step is carried out according to a method known per se, for example, the method described in Jikken Kagaku Kouza, 5th Edition, vol. 13-19 (the Chemical Society of Japan ed.); Shin Jikken Kagaku Kouza, vol. 14-15 (the Chemical Society of Japan ed.); Fine Organic Chemistry, Revised 2nd Edition (L. F. Fieser, Th. Eicher, Nankodo); Organic Name Reactions, the Reaction Mechanism and Essence, Revised Edition (Hideo Togo, Kodansha); ORGANIC SYNTHESSES Collective Volume I-VII (John Wiley & Sons Inc.); Modern Organic Synthesis in the Laboratory A Collection of Standard Experimental Procedures (Jie Jack Li, OXFORD UNIVERSITY); Comprehensive Heterocyclic Chemistry III, Vol. 1-Vol. 14 (Elsevier Japan); Strategic Applications of Named Reactions in Organic Synthesis (translated by Kiyoshi Tomioka, Kagakudojin); Comprehensive Organic Transformations (VCH Publishers Inc.), 1989, or the like, or the method described in Examples.

(70) In each step, the protection or deprotection reaction of a functional group is carried out according to a method known per se, for example, the method described in "Protective Groups in Organic Synthesis, 4th Ed", Wiley-Interscience, Inc., 2007 (Theodora W. Greene, Peter G. M. Wuts); "Protecting Groups 3rd Ed." Thieme, 2004 (P. J. Kocienski), or the like, or the method described in Examples.

(71) Examples of the protecting group for a hydroxy group of an alcohol and the like and a phenolic hydroxy group include ether-type protecting groups such as methoxymethyl ether, benzyl ether, tert-butyldimethylsilyl ether, tetrahydropyranyl ether and the like; carboxylate ester-type protecting groups such as acetate ester and the like; sulfonate ester-type protecting groups such as methanesulfonate ester and the like; carbonate ester-type protecting groups such as tert-butylcarbonate and the like, and the like.

(72) Examples of the protecting group for a carbonyl group of an aldehyde include acetal-type protecting groups such as dimethylacetal and the like; cyclic acetal-type protecting groups such as 1,3-dioxane and the like, and the like.

(73) Examples of the protecting group for a carbonyl group of a ketone include ketal-type protecting groups such as dimethylketal and the like; cyclic ketal-type protecting groups such as 1,3-dioxane and the like; oxime-type protecting groups such as O-methyloxime and the like; hydrazone-type protecting groups such as N,N-dimethylhydrazone and the like, and the like.

(74) Examples of the protecting group for a carboxyl group include ester-type protecting groups such as methyl ester and the like; amide-type protecting groups such as N,N-dimethylamide and the like, and the like.

(75) Examples of the protecting group for a thiol include ether-type protecting groups such as benzyl thioether and the like; ester-type protecting groups such as thioacetate ester, thiocarbonate, thiocarbamate and the like, and the like.

(76) Examples of the protecting group for an amino group and an aromatic heterocycle such as imidazole, pyrrole, indole and the like include carbamate-type protecting groups such as benzyl carbamate and the like; amide-type protecting groups such as acetamide and the like; alkyl amine-type protecting groups such as N-triphenylmethylamine and the like; sulfonamide-type protecting

groups such as methanesulfonamide and the like, and the like.

(77) The protecting groups can be removed according to a method known per se, for example, by employing a method using acid, base, ultraviolet rays, hydrazine, phenylhydrazine, sodium N-methyldithiocarbamate, tetrabutylammonium fluoride, palladium acetate, trialkylsilyl halide (e.g., trimethylsilyl iodide, trimethylsilyl bromide) and the like, a reduction method, and the like.

(78) When reduction reaction is carried out in each step, examples of the reducing agent to be used include metal hydrides such as lithium aluminium hydride, sodium triacetoxyborohydride, sodium cyanoborohydride, diisobutylaluminium hydride (DIBAL-H), sodium borohydride, tetramethylammonium triacetoxyborohydride and the like; boranes such as borane tetrahydrofuran complex and the like; Raney nickel; Raney cobalt; hydrogen; formic acid; triethylsilane and the like. When carbon-carbon double bond or triple bond is reduced, a method using a catalyst such as palladium-carbon, Lindlar's catalyst and the like may be employed.

(79) When oxidation reaction is carried out in each step, examples of the oxidizing agent to be used include peroxides such as m-chloroperbenzoic acid (mCPBA), hydrogen peroxide, tert-butylhydroperoxide and the like; perchlorates such as tetrabutylammonium perchlorate and the like; chlorates such as sodium chlorate and the like; chlorites such as sodium chlorite and the like; periodates such as sodium periodate and the like; hypervalent iodine reagents such as iodosylbenzene and the like; reagents containing manganese such as manganese dioxide, potassium permanganate and the like; leads such as lead tetraacetate and the like; reagents containing chromium such as pyridinium chlorochromate (PCC), pyridinium dichromate (PDC), Jones reagent and the like; halogen compounds such as N-bromosuccinimide (NBS) and the like; oxygen; ozone; sulfur trioxide-pyridine complex; osmium tetroxide; selenium dioxide; 2,3-dichloro-5,6-dicyano-1,4-benzoquinone (DDQ) and the like.

(80) When radical cyclization reaction is carried out in each step, examples of the radical initiator to be used include azo compounds such as azobisisobutyronitrile (AIBN) and the like; water-soluble radical initiators such as 4-4'-azobis-4-cyanopentanoic acid (ACPA) and the like; triethylboron in the presence of air or oxygen; benzoyl peroxide and the like. Examples of the radical reagent to be used include tributylstannane, tris(trimethylsilyl)silane, 1,1,2,2-tetraphenyldisilane, diphenylsilane, samarium iodide and the like.

(81) When Wittig reaction is carried out in each step, examples of the Wittig reagent to be used include alkylidene phosphoranes and the like. The alkylidene phosphoranes can be prepared according to a method known per se, for example, by reacting a phosphonium salt with a strong base.

(82) When Horner-Emmons reaction is carried out in each step, examples of the reagent to be used include phosphonoacetates such as methyl dimethylphosphonoacetate, ethyl diethylphosphonoacetate and the like; and bases such as alkali metal hydrides, organic lithiums and the like.

(83) When Friedel-Crafts reaction is carried out in each step, examples of the reagent to be used include a combination of a Lewis acid and an acid chloride or a combination of a Lewis acid and an alkylating agent (e.g., an alkyl halide, an alcohol, an olefin etc.). Alternatively, an organic acid or an inorganic acid can also be used instead of a Lewis acid, and an anhydride such as acetic anhydride and the like can also be used instead of an acid chloride.

(84) When aromatic nucleophilic substitution reaction is carried out in each step, a nucleophile (e.g., an amine, imidazole etc.) and a base (e.g., an organic base etc.) are used as reagents.

(85) When nucleophilic addition reaction by a carbo anion, nucleophilic 1,4-addition reaction (Michael addition reaction) by a carbo anion or nucleophilic substitution reaction by a carbo anion is carried out in each step, examples of the base to be used for generation of the carbo anion include organic lithiums, metal alkoxides, inorganic bases, organic bases and the like.

(86) When Grignard reaction is carried out in each step, examples of the Grignard reagent to be used include arylmagnesium halides such as phenylmagnesium bromide and the like; and

alkylmagnesium halides such as methylmagnesium bromide and the like. The Grignard reagent can be prepared according to a method known per se, for example, by reacting an alkyl halide or an aryl halide with a metal magnesium or isopropylmagnesium chloride-lithium chloride complex in an ether, tetrahydrofuran and the like as a solvent.

(87) When Knoevenagel condensation reaction is carried out in each step, a compound having an activated methylene group with two electron withdrawing groups (e.g., malonic acid, diethyl malonate, malononitrile etc.) and a base (e.g., an organic base, a metal alkoxide, an inorganic base) are used as a reagent.

(88) When Vilsmeier-Haack reaction is carried out in each step, phosphoryl chloride and an amide derivative (e.g., N,N-dimethylformamide etc.) are used as a reagent.

(89) When azidation reaction of an alcohol, an alkyl halide or a sulfonate is carried out in each step, examples of the azidating agent to be used include diphenylphosphorylazide (DPPA), trimethylsilylazide, sodium azide and the like. For example, for the azidation reaction of an alcohol, a method using diphenylphosphorylazide and 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU), a method using trimethylsilylazide and a Lewis acid, a method using diphenylphosphorylazide, triphenylphosphine and azodicarboxylate, and the like are employed.

(90) When reductive amination reaction is carried out in each step, examples of the reducing agent to be used include sodium triacetoxyborohydride, sodium cyanoborohydride, hydrogen, formic acid and the like. When the substrate is an amine compound, examples of the carbonyl compound to be used include paraformaldehyde, aldehydes such as acetaldehyde and the like, and ketones such as cyclohexanone and the like. When the substrate is a carbonyl compound, examples of the amine to be used include ammonia, primary amines such as methylamine and the like; secondary amines such as dimethylamine and the like, and the like.

(91) When Mitsunobu reaction is carried out in each step, an azodicarboxylate (e.g., diethyl azodicarboxylate (DEAD), diisopropyl azodicarboxylate (DIAD) etc.) and triphenylphosphine are used as a reagent.

(92) When esterification reaction, amidation reaction or ureation reaction is carried out in each step, examples of the reagent to be used include acyl halides such as acid chlorides, acid bromides and the like; activated carboxylic acids such as anhydrides, activated esters, sulfates and the like. Examples of the activating agent of the carboxylic acid include carbodiimide condensing agents such as 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide hydrochloride (EDC HCl) and the like; triazine condensing agents such as 4-(4,6-dimethoxy-1,3,5-triazin-2-yl)-4-methylmorpholinium chloride n-hydrate (DMT-MM) and the like; carbonate condensing agents such as 1,1-carbonyldiimidazole (CDI) and the like; diphenylphosphorylazide (DPPA); benzotriazol-1-yloxy-trisdimethylaminophosphonium salt (BOP reagent); 2-chloro-1-methyl-pyridinium iodide (Mukaiyama reagent); thionyl chloride; lower alkyl haloformates such as ethyl chloroformate and the like; O-(7-azabenzotriazol-1-yl)-N,N,N',N'-tetramethyluronium hexafluorophosphate (HATU); sulfuric acid; combinations thereof and the like. When carbodiimide condensing agent is used, an additive such as 1-hydroxybenzotriazole (HOBt), N-hydroxysuccinimide (HOSu), dimethylaminopyridine (DMAP) and the like may be added to the reaction system.

(93) When coupling reaction is carried out in each step, examples of the metal catalyst to be used include palladium compounds such as palladium(II) acetate, tetrakis(triphenylphosphine)palladium(0), dichlorobis(triphenylphosphine)palladium(II), dichlorobis(triethylphosphine)palladium(II), tris(dibenzylideneacetone)dipalladium(0), 1,1'-bis(diphenylphosphino)ferrocene palladium(II) chloride and the like; nickel compounds such as tetrakis(triphenylphosphine)nickel(0) and the like; rhodium compounds such as tris(triphenylphosphine)rhodium(III) chloride and the like; cobalt compounds; copper compounds such as copper oxide, copper(I) iodide and the like; platinum compounds and the like. In addition, a base can be added to the reaction system, and examples thereof include inorganic bases and the like.

(94) When thiocarbonylation reaction is carried out in each step, phosphorus pentasulfide is typically used as the thiocarbonylating agent. Alternatively, a reagent having a 1,3,2,4-dithiadiphosphetane-2,4-disulfide structure (e.g., 2,4-bis(4-methoxyphenyl)-1,3,2,4-dithiadiphosphetane-2,4-disulfide (Lawesson reagent) etc.) can also be used instead of phosphorus pentasulfide.

(95) When halogenation reaction is carried out in each step, examples of the halogenating agent to be used include N-iodosuccinimide, N-bromosuccinimide (NBS), N-chlorosuccinimide (NCS), bromine, sulfonyl chloride, trichloroisocyanuric acid and the like. In addition, the reaction can be accelerated by subjecting a radical initiator such as heat, light, benzoyl peroxide, azobisisobutyronitrile and the like to the reaction system reaction.

(96) When halogenation reaction of a hydroxy group is carried out in each step, examples of the halogenating agent to be used include hydrohalic acids and acid halides of inorganic acids, specifically, hydrochloric acid, thionyl chloride, phosphorus oxychloride and the like for chlorination, 48% hydrobromic acid and the like for bromination. In addition, a method of producing an alkyl halide by reacting an alcohol with triphenylphosphine and carbon tetrachloride or carbon tetrabromide or the like can be employed. Alternatively, a method of producing an alkyl halide via two steps comprising converting an alcohol to the corresponding sulfonate, and then reacting the sulfonate with lithium bromide, lithium chloride or sodium iodide can also be employed.

(97) When Arbuzov reaction is carried out in each step, examples of the reagent to be used include alkyl halides such as ethyl bromoacetate and the like; and phosphites such as triethyl phosphite, tri(isopropyl) phosphite and the like.

(98) When sulfonate esterification reaction is carried out in each step, examples of the sulfonating agent to be used include methanesulfonyl chloride, p-toluenesulfonyl chloride, methanesulfonic anhydride, p-toluenesulfonic anhydride and the like.

(99) When hydrolysis reaction is carried out in each step, an acid or a base is used as a reagent. For acid hydrolysis reaction of tert-butyl ester, formic acid, triethylsilane and the like may be added to reductively-trap tert-butyl cation which is by-produced.

(100) When dehydration reaction is carried out in each step, examples of the dehydrating agent to be used include sulfuric acid, diphosphorus pentoxide, phosphorus oxychloride, N,N'-dicyclohexylcarbodiimide, alumina, polyphosphoric acid and the like.

(101) When alkylation reaction is carried out in each step, an electrophile (e.g., an alkyl halide etc.) and a base (e.g., an organic base, an inorganic base, a metal alkoxide, a metal amide etc.) are used as reagents.

(102) When cyanidation reaction is carried out in each step, examples of the reagent to be used include metal cyanides such as copper cyanide, potassium cyanide and the like.

(103) Compound (I) can be produced according to the production methods shown in the following Scheme 1 or Scheme 2. Each symbol in the formulas of the schemes is as defined above, unless otherwise specified.

(104) Moreover, compound (I) can be produced by carrying out protection reaction, deprotection reaction, amidation reaction, sulfonamidation reaction, ureation reaction, carbamoylation reaction, alkylation reaction, Mitsunobu reaction, hydrogenation reaction, oxidation reaction, reduction reaction, halogenation reaction, coupling reaction, nucleophilic addition reaction by a carbo anion, Grignard reaction, deoxofluorination reaction, dehydration reaction and the like singly or two or more thereof in combination.

(105) The below-mentioned compound (Ia) (compound (I) wherein R^{sup.1} is R^{sup.1a}) can be produced according to the method shown in the following Scheme 1. R^{sup.1a} is an optionally substituted cyclopropyl group, an optionally substituted C_{sub.6-14} aryl group, or an optionally substituted 5- or 6-membered monocyclic aromatic heterocyclic group. The other symbols are as defined above.

(106) ##STR00017## ##STR00018##

(107) Compound (4) can be produced by subjecting compound (2) to a coupling reaction with compound (3) using a palladium catalyst and a base under carbon monoxide atmosphere. Examples of the palladium catalyst include a combination of 4,5-bis(diphenylphosphino)-9,9-dimethylxanthene (alias: Xantphos) and bis(dibenzylideneacetone)palladium(0), and the like. Examples of the base include N,N-dicyclohexylmethylamine.

(108) Compound (7) can be produced by subjecting compound (6) to a dehydration cyclization reaction. A combination of Burgess reagent or p-toluenesulfonyl chloride and a base is used as a reagent. Examples of the base include the above-mentioned organic bases (e.g., N,N-diisopropylethylamine, triethylamine, etc.).

(109) Compound (12) can be produced by subjecting compound (9), compound (10) and compound (11) to a three-component condensation reaction. Examples of the reagent to be used include formic acid.

(110) Compound (14) can be produced by subjecting compound (12) to the Stetter reaction with compound (13). A combination of 3-ethyl-5-(2-hydroxyethyl)-4-methylthiazolium bromide and a base is used as a reagent. Examples of the base include triethylamine.

(111) Compound (15) can be produced by subjecting compound (14) to a reduction reaction. Examples of the reagent to be used include sodium borohydride and potassium tri(sec-butyl)borohydride. With regard to the ratio of stereoisomers of the product, only the anti-form (15a) shown below is obtained, or a mixture of the anti-form (15a) and the syn-form (15b) shown below is obtained (the stereochemistry of the formula is relative configuration).

(112) ##STR00019##

(113) Compound (Ia) can be produced by subjecting compound (16) to an alkylation reaction with compound (8), followed by a cyclization reaction. The cyclization reaction follows the alkylation reaction, but it may be carried out in stages. In this case, the cyclization reaction can be carried out under basic or acidic condition. Examples of the base include triethylamine, DIPEA, potassium acetate and the like. Examples of the acid to be used include acetic acid and the like.

(114) The below-mentioned compound (Ib) (compound (I) wherein R^{sup.1} is R^{sup.1b}) can be produced from compound (17) according to the method shown in the following Scheme 2. R^{sup.1b} is an optionally substituted cycloalkyl group, an optionally substituted C_{sub.6-14} aryl group, an optionally substituted 5- or 6-membered monocyclic aromatic heterocyclic group, or an optionally substituted 3- to 8-membered monocyclic non-aromatic heterocyclic group. X is a halogen atom. The other symbols are as defined above.

(115) ##STR00020##

(116) Compound (23) can be produced by subjecting compound (22) to a reduction reaction. Examples of the reagent to be used include sodium borohydride and potassium tri(sec-butyl)borohydride. Alternatively, a combination of aluminium triisopropoxide and 2-propanol may also be used. With regard to the ratio of stereoisomers of the product, only the anti-form (23a) shown below is obtained, or a mixture of the anti-form (23a) and the syn-form (23b) shown below is obtained (the stereochemistry of the formula is relative configuration).

(117) ##STR00021##

(118) Compound (Ib) can be produced by subjecting compound (24) to an alkylation reaction with compound (8), followed by a cyclization reaction. The cyclization reaction follows the alkylation reaction, but it may be carried out in stages. In this case, the cyclization reaction can be carried out under basic or acidic condition. Examples of the base include triethylamine, DIPEA, potassium acetate and the like. Examples of the acid to be used include acetic acid and the like.

(119) Compounds (2), (3), (9), (10), (11), (13), (17), (18), (20) and (21) which are used as raw materials in each production method may be commercially easily available or can be produced according to a method known per se.

(120) As for the configurational isomers (E, Z forms) of compound (I), they can be isolated and

purified when isomerization occurs, for example, according to a conventional separation means such as extraction, recrystallization, distillation, chromatography and the like to obtain a pure compound. In addition, the corresponding pure isomer can also be obtained by isomerizing a double bond using heating, an acid catalyst, a transition metal complex, a metal catalyst, a radical catalyst, light irradiation, a strong base catalyst and the like, according to the method described in Shin Jikken Kagaku Kouza 14 (The Chemical Society of Japan ed.), pages 251 to 253, or 4th Edition Jikken Kagaku Kouza 19 (The Chemical Society of Japan ed.), pages 273 to 274, or a method analogous thereto.

(121) Compound (I) contains a stereoisomer depending on the kind of a substituent, and each stereoisomer and a mixture thereof are encompassed in the present invention.

(122) Compound (I) may be a hydrate or a non-hydrate.

(123) When the objective product is obtained as a free form by the above-mentioned reaction, it can be converted to a salt according to a conventional method, or when the objective product is obtained as a salt, it can be converted to a free form or other salt according to a conventional method. The thus-obtained compound (I) can also be isolated and purified from a reaction mixture according to a known method such as phase transfer, concentration, solvent extraction, distillation, crystallization, recrystallization, chromatography and the like.

(124) When compound (I) contains a configurational isomer, a diastereomer, a conformer and the like, each can be isolated according to the above-mentioned separation and purification methods, if desired. In addition, when compound (I) is racemic, d-form and l-form can be isolated according to a conventional optical resolution such as preparative high performance liquid chromatography (preparative HPLC), supercritical fluid chromatography (preparative SFC) and the like.

(125) The thus-obtained compound (I), other reaction intermediate therefor and starting compounds thereof can be isolated and purified from a reaction mixture according to a method known per se, for example, extraction, concentration, neutralization, filtration, distillation, recrystallization, column chromatography, thin layer chromatography, preparative high performance liquid chromatography (preparative HPLC), moderate-pressure preparative liquid chromatography (moderate-pressure preparative LC) and the like.

(126) A salt of compound (I) can be produced according to a method known per se. For example, when compound (I) is a basic compound, it can be produced by adding an inorganic acid or organic acid, or when compound (I) is an acidic compound, by adding an organic base or inorganic base.

(127) When compound (I) contains an optical isomer, each optical isomer and a mixture thereof are encompassed in the scope of the present invention, and these isomers can be subjected to optical resolution or can be produced respectively, according to a method known per se, if desired.

(128) Compound (I) may be a crystal.

(129) The crystal of compound (I) can be produced according to a crystallization method known per se.

(130) Examples of the crystallization method include crystallization method from a solution, crystallization method from vapor, crystallization method from a melt, and the like.

(131) The "crystallization method from a solution" is typically a method of shifting a non-saturated state to supersaturated state by varying factors involved in solubility of compounds (solvent composition, pH, temperature, ionic strength, redox state, etc.) or the amount of solvent. Specific examples thereof include a concentration method, a slow cooling method, a reaction method (a diffusion method, an electrolysis method), a hydrothermal growth method, a flux method and the like. Examples of the solvent to be used include aromatic hydrocarbons (e.g., benzene, toluene, xylene, etc.), halogenated hydrocarbons (e.g., dichloromethane, chloroform, etc.), saturated hydrocarbons (e.g., hexane, heptane, cyclohexane, etc.), ethers (e.g., diethyl ether, diisopropyl ether, tetrahydrofuran, dioxane, etc.), nitriles (e.g., acetonitrile, etc.), ketones (e.g., acetone, etc.), sulfoxides (e.g., dimethyl sulfoxide, etc.), acid amides (e.g., N,N-dimethylformamide, etc.), esters (e.g., ethyl acetate, isopropyl acetate, etc.), alcohols (e.g., methanol, ethanol, 2-propanol, etc.),

water and the like. These solvents are used alone or in a combination of two or more at a suitable ratio (e.g., 1:1 to 1:100 (a volume ratio)). Where necessary, a seed crystal can be used.

(132) The “crystallization method from vapor” is, for example, a vaporization method (a sealed tube method, a gas stream method), a gas phase reaction method, a chemical transportation method and the like.

(133) The “crystallization method from a melt” is, for example, a normal freezing method (a pulling method, a temperature gradient method, a Bridgman method), a zone melting method (a zone leveling method, a floating zone method), a special growth method (a VLS method, a liquid phase epitaxy method) and the like.

(134) Preferable examples of the crystallization method include a method comprising dissolving compound (I) in a suitable solvent (e.g., alcohols such as methanol, ethanol etc.) at 20° C. to 120° C., and cooling the obtained solution to a temperature (e.g., 0 to 50° C., preferably 0 to 20° C.) not higher than the dissolution temperature, and the like.

(135) The thus-obtained crystals of the present invention can be isolated, for example, by filtration and the like.

(136) An analysis method of the obtained crystal is generally a method of crystal analysis by powder X-ray diffraction. As a method of determining crystal orientation, a mechanical method or an optical method and the like can also be used.

(137) The crystal of compound (I) obtained by the above-mentioned production method may have high purity, high quality, and low hygroscopicity, may not be denatured even after a long-term preservation under general conditions, and may be expected to be superior in the stability. In addition, it may be also superior in the biological properties (e.g., pharmacokinetics (absorption, distribution, metabolism, excretion), efficacy expression etc.) and may be extremely useful as a medicament.

(138) Compound (I) may be a prodrug. The prodrug of compound (I) means a compound which is converted to compound (I) with a reaction due to an enzyme, gastric acid and the like under the physiological condition in the living body, that is, a compound which is converted to compound (I) by enzymatic oxidation, reduction, hydrolysis and the like; a compound which is converted to compound (I) by hydrolysis and the like due to gastric acid, and the like. Examples of the prodrug for compound (I) include a compound obtained by subjecting an amino group in compound (I) to acylation, alkylation or phosphorylation (e.g., a compound obtained by subjecting an amino group in compound (I) to eicosanoylation, alanylation, pentylaminocarbonylation, (5-methyl-2-oxo-1,3-dioxolen-4-yl)methoxycarbonylation, tetrahydrofurylation, pyrrolidylmethylation, pivaloyloxymethylation or tert-butylation); a compound obtained by subjecting a hydroxy group in compound (I) to acylation, alkylation, phosphorylation or boration (e.g., a compound obtained by subjecting a hydroxy group in compound (I) to acetylation, palmitoylation, propanoylation, pivaloylation, succinylation, fumarylation, alanylation or dimethylaminomethylcarbonylation); a compound obtained by subjecting a carboxyl group in compound (I) to esterification or amidation (e.g., a compound obtained by subjecting a carboxyl group in compound (I) to ethyl esterification, phenyl esterification, carboxymethyl esterification, dimethylaminomethyl esterification, pivaloyloxymethyl esterification, ethoxycarbonyloxyethyl esterification, phthalidyl esterification, (5-methyl-2-oxo-1,3-dioxolen-4-yl)methyl esterification, cyclohexyloxycarbonyl ethyl esterification or methylamidation) and the like. These compounds can be produced from compound (I) according to a method known per se.

(139) The prodrug of compound (I) may also be one which is converted to compound (I) under physiological conditions as described in “IYAKUHIN no KAIHATSU (Development of Pharmaceuticals)”, Vol. 7, Design of Molecules, p. 163-198, Published by HIROKAWA SHOTEN (1990).

(140) In the present specification, compound (I) and the prodrug of compound (I) are sometimes collectively abbreviated as “the compound of the present invention”.

(141) Compound (I) may be a hydrate, a non-hydrate, a solvate or a non-solvate.

(142) In addition, compound (I) may be a compound labeled or substituted with an isotope (e.g., ²H, ³H, ¹¹C, ¹⁴C, ¹⁸F, ³⁵S, ¹²⁵I) and the like. The compound labeled or substituted with an isotope may be used, for example, as a tracer (PET tracer) used in positron emission tomography (PET), and useful in the field of medical diagnosis and the like.

(143) Compound (I) also encompasses a deuterium conversion form wherein ¹H is converted to ²H(D).

(144) Compound (I) also encompasses a tautomer thereof.

(145) Compound (I) may be a pharmaceutically acceptable cocrystal or a salt thereof. The cocrystal or a salt thereof means a crystalline substance constituted with two or more special solids at room temperature, each having different physical properties (e.g., structure, melting point, melting heat, hygroscopicity, solubility and stability). The cocrystal or a salt thereof can be produced according to a cocrystallization a method known per se.

(146) Compound (I) may also be used as a PET tracer.

(147) Since the compound of the present invention has a superior HDAC inhibitory action, preferably class II HDAC inhibitory action, more preferably HDAC6 inhibitory action, it may be also useful as safe medicaments based on such action.

(148) For example, the medicament of the present invention containing the compound of the present invention may be expected to show low toxicity (e.g., acute toxicity, chronic toxicity, genetic toxicity, hematotoxicity, reproductive toxicity, cardiotoxicity, carcinogenicity), and used for a mammal (e.g., mouse, rat, hamster, rabbit, cat, dog, bovine, sheep, monkey, human) as a prophylactic or therapeutic agent for HDAC-associated diseases, preferably class II HDAC-associated diseases, more preferably HDAC6-associated diseases, more specifically, the diseases described in (1)-(7) below.

(149) Particularly, the compound of the present invention may be expected to show low genetic toxicity, and therefore, the medicament of the present invention may be expected to show low genetic toxicity. (1) inflammatory diseases (e.g., acute pancreatitis, chronic pancreatitis, asthma, adult respiratory distress syndrome, chronic obstructive pulmonary disease (COPD), idiopathic pulmonary fibrosis, inflammatory bone disease, inflammatory pulmonary disease, inflammatory bowel disease, celiac disease, hepatitis, systemic inflammatory response syndrome (SIRS), postoperative or posttraumatic inflammation, pneumonia, nephritis, meningitis, cystitis, pharyngolaryngitis, gastric mucosal injury, spondylitis, arthritis, dermatitis, chronic pneumonia, bronchitis, pulmonary infarction, silicosis, pulmonary sarcoidosis, diabetic nephropathy, uveitis, suppurative hidradenitis etc.), (2) autoimmune diseases (e.g., rheumatoid arthritis, psoriasis, inflammatory bowel disease (e.g., Crohn's disease, ulcerative colitis etc.), Sjogren's syndrome, Behcet's disease, multiple sclerosis, systemic lupus erythematosus, lupus nephritis, discoid lupus erythematosus, Castleman's disease, ankylopoietic spondylarthritis, polymyositis, dermatomyositis (DM), polyarteritis nodosa (PN), mixed connective tissue disease (MCTD), scleroderma, profundus lupus erythematosus, chronic thyroiditis, Graves' disease, autoimmune gastritis, type I diabetes, autoimmune hemolytic anemia, autoimmune neutropenia, thrombocytopenia, atopic dermatitis, pemphigus, chronic active hepatitis, myasthenia gravis, graft versus host disease, Addison's disease, abnormal immunoresponse, arthritis, dermatitis, radiodermatitis, primary biliary cirrhosis etc.), (3) osteoarticular degenerative disease (e.g., rheumatoid arthritis, osteoporosis, osteoarthritis etc.), (4) neoplastic diseases [e.g., malignant tumor, angiogenesis glaucoma, infantile hemangioma, multiple myeloma, chronic sarcoma, metastasis melanoma, Kaposi's sacroma, vascular proliferation, cachexia, metastasis of the breast cancer, cancer (e.g., colorectal cancer (e.g., familial colorectal cancer, hereditary nonpolyposis colorectal cancer, gastrointestinal stromal tumor etc.), lung cancer (e.g., non-small cell lung cancer, small cell lung cancer, malignant mesothelioma etc.), mesothelioma, pancreatic cancer (e.g., pancreatic duct cancer etc.), gastric cancer (e.g., papillary adenocarcinoma, mucinous adenocarcinoma, adenosquamous carcinoma, etc.), breast cancer (e.g.,

invasive ductal carcinoma, ductal carcinoma in situ, inflammatory breast cancer etc.), ovarian cancer (e.g., ovarian epithelial carcinoma, extragonadal germ cell tumor, ovarian germ cell tumor, ovarian low malignant potential tumor etc.), prostate cancer (e.g., hormone-dependent prostate cancer, non-hormone dependent prostate cancer etc.), liver cancer (e.g., primary liver cancer, extrahepatic bile duct cancer etc.), thyroid cancer (e.g., medullary thyroid carcinoma etc.), kidney cancer (e.g., renal cell carcinoma, transitional cell carcinoma in kidney and urinary duct etc.), uterine cancer, brain tumor (e.g., pineal astrocytoma, pilocytic astrocytoma, diffuse astrocytoma, anaplastic astrocytoma etc.), melanoma, sarcoma, urinary bladder cancer, hematologic cancer and the like including multiple myeloma, hypophyseal adenoma, glioma, acoustic neurinoma, retinoblastoma, pharyngeal cancer, laryngeal cancer, cancer of the tongue, thymoma, esophagus cancer, duodenal cancer, colorectal cancer, rectal cancer, hepatoma, pancreatic endocrine tumor, bile duct cancer, gallbladder cancer, penile cancer, urinary duct cancer, testis tumor, vulvar cancer, cervix cancer, endometrial cancer, uterus sarcoma, cholesteric disease, vaginal cancer, skin cancer, fungoid mycosis, basal cell tumor, soft tissue sarcoma, malignant lymphoma, Hodgkin's disease, myelodysplastic syndrome, adult T cell leukemia, chronic bone marrow proliferative disease, pancreatic endocrine, tumor fibrous histiocytoma, leiomyosarcoma, rhabdomyosarcoma, cancer of unknown primary), leukemia (e.g., acute leukemia (e.g., acute lymphatic leukemia, acute myelocytic leukemia etc.), chronic leukemia (e.g., chronic lymphatic leukemia, chronic myelocytic leukemia etc.), myelodysplastic syndrome), uterine sarcoma (e.g., mixed mesodermal tumor, uterine leiomyosarcoma, endometrial stromal tumor etc.), myelofibrosis etc.], (5) neurodegenerative diseases and/or central diseases (i) psychiatric diseases [e.g., depression, major depression, bipolar depression, dysthymic disorder, emotional disorder (seasonal affective disorder and the like), recurrent depression, postpartum depression, stress disorder, depression symptom, mania, anxiety, generalized anxiety disorder, anxiety syndrome, panic disorder, phobia, social phobia, social anxiety disorder, obsessive disorder, post-traumatic stress syndrome, post-traumatic stress disorder, Tourette syndrome, autism, autistic spectrum syndrome, fragile X syndrome, Rett syndrome, adjustment disorder, bipolar disorder, neurosis, schizophrenia (e.g., positive symptom, negative symptom, cognitive symptom), cognitive dysfunction associated with schizophrenia, chronic fatigue syndrome, anxiety neurosis, compulsive neurosis, epilepsy, anxiety, anxious mental state, emotional abnormality, cyclothymia, nervous erethism, faint, addiction, low sex drive, attention deficit hyperactivity disorder (ADHD), psychotic major depression, refractory major depression, treatment-resistant depression, hereditary spastic paraplegia], (ii) neurodegenerative diseases [e.g., Alzheimer's disease, dementia of Alzheimer type, Alzheimer-type senile dementia, Parkinson's disease, muscular dystrophy, Parkinson's disease associated with dementia, Huntington's disease, multi-infarct dementia, frontotemporal lobar degeneration [progressive supranuclear palsy, corticobasal degeneration, frontotemporal dementia and parkinsonism linked to MAPT mutation (FTDP-17), frontotemporal dementia, Pick's disease, argyrophilic grain dementia etc.], Parkinson's type dementia, Niemann-Pick syndrome, Down's disease, vascular dementia, postencephalitic parkinsonism, Lewy body dementia, Rubinstein-Taybi syndrome, HIV dementia, amyotrophic lateral sclerosis (ALS), motor neurogenesis disease (MND), Creutzfeldt-Jakob disease or prion disease, cerebral palsy, multiple sclerosis, Riley-Day syndrome], (iii) age-related cognition memory disorders [e.g., age-related memory disorders, senile dementia], (iv) sleep disorders [e.g., intrinsic sleep disorders (e.g., psychophysiological insomnia and the like), extrinsic sleep disorder, circadian rhythm disorders (e.g., time zone change syndrome (jet lag), shift work sleep disorder, irregular sleep-wake pattern, delayed sleep phase syndrome, advanced sleep phase syndrome, non-24-hour sleep-wake and the like), parasomnia, sleep disorders associated with internal medical or psychiatric disorder (e.g., chronic obstructive pulmonary diseases, Alzheimer's disease, Parkinson's disease, cerebrovascular dementia, schizophrenia, depression, anxiety neurosis), stress insomnia, insomnia, insomniac neurosis, sleep apnea syndrome], (v) respiratory depression caused by anesthetics, traumatic disease, or neurodegenerative disease and the like, (vi) traumatic brain injury,

cerebral apoplexy, neurotic anorexia, eating disorder, anorexia nervosa, hyperorexia, other eating disorder, alcohol dependence, alcohol abuse, alcoholic amnesia, alcohol paranoia, alcohol preference, alcohol withdrawal, alcoholic insanity, alcohol poisoning, alcoholic jealousy, alcoholic mania, alcohol-dependent psychiatric disorder, alcoholic insanity, pharmacophilia, pharmacophobia, pharmacomania, drug withdrawal, migraine, stress headache, catatonic headache, diabetic neuropathy, obesity, diabetes, muscular spasm, Meniere's disease, autonomic ataxia, alopecia, glaucoma, hypertension, cardiac disease, tachycardia, congestive cardiac failure, hyperventilation, bronchial asthma, apnea, sudden infant death syndrome, inflammatory disease, allergic disease, impotence, climacteric disorder, infertility, cancer, immunodeficiency syndrome caused by HIV infection, immunodeficiency syndrome caused by stress, cerebrospinal meningitis, acromegaly, incontinence, metabolic syndrome, osteoporosis, peptic ulcer, irritable bowel syndrome, inflammatory bowel disease, ulcerative colitis, Crohn's disease, stress gastrointestinal disorder, stress vomiting, peptic ulcer, diarrhea, constipation, postoperative ileus, (vii) pain (pain, cancer pain, acute pain caused by inflammation, pain associated with chronic inflammation, postoperative pain (incision pain, deep pain, visceral pain, chronic pain after operation and the like), muscular pain (muscular pain associated with chronic pain disease, stiff shoulder and the like), arthralgia, toothache, temporomandibular joint pain, headache (migraine, catatonic headache, headache associated with fever, headache associated with hypertension), visceral pain (cardiac pain, angina pain, abdominal pain, renal pain, urinary tract pain, bladder pain), obstetric and gynecologic pain (mittelschmerz, dysmenorrhea, labor pain), neuropathic pain (hernia of intervertebral disk, nerve root pain, neuralgia after herpes zoster, trigeminal neuralgia, lumbago and the like), peripheral neuropathy (CIPN) derived from anticancer drugs (taxane anti-cancer agent (e.g., paclitaxel (taxol), docetaxel), vinca alkaloid anti-cancer agent (e.g., vincristine, vinblastine), platinum preparation (e.g., cisplatin, carboplatin, oxaliplatin), molecularly targeted drug (e.g., bortezomib) and the like) and neurological symptoms associated therewith (chemotherapy-induced neuropathic pain (CINP))), (6) chronic heart failure or acute heart failure, acute decompensated heart failure, ischemic heart disease, cardiomyopathy, myocarditis, valvular disease, (7) peripheral neuropathy and the like (e.g., demyelinating diseases and neuropathy (multiple sclerosis, Guillain-Barre syndrome, Fisher syndrome, chronic inflammatory demyelinating polyneuropathy (CIDP), multifocal motor neuropathy (MMN), Charcot-Marie-Tooth disease, hereditary sensory and autonomic neuropathy, familial amyloidotic polyneuropathy)).

(150) The medicament of the present invention may be preferably used as an agent for the prophylaxis or treatment of autoimmune disease, inflammatory disease, osteoarticular degenerative disease, neurodegenerative disease, central disease, neoplastic disease, or peripheral neuropathy, more preferably inflammatory bowel disease (preferably Crohn's disease or ulcerative colitis), systemic lupus erythematosus, rheumatoid arthritis, psoriasis, Sjogren's syndrome, Behcet's syndrome, multiple sclerosis, graft versus host disease, Alzheimer's disease (preferably dementia of Alzheimer type), schizophrenia, dementia with Lewy Bodies, frontotemporal lobar degeneration [progressive supranuclear palsy, corticobasal degeneration, frontotemporal dementia and parkinsonism linked to MAPT mutation (FTDP-17), frontotemporal dementia, Pick's disease, argyrophilic grain dementia etc.], Parkinson's disease, Huntington's disease, Rubinstein-Taybi Syndrome, muscular dystrophy, Rett Syndrome, amyotrophic lateral sclerosis, Charcot-Marie-Tooth disease, depression, hereditary spastic paraplegia, Riley-Day syndrome, Castleman's disease, leukemia, uterine leiomyosarcoma, prostate cancer, colon cancer, multiple myeloma, cachexia or myelofibrosis, chronic heart failure or acute heart failure, acute decompensated heart failure, ischemic heart disease, cardiomyopathy, myocarditis, valvular disease, peripheral neuropathy and the like.

(151) The medicament of the present invention may be more preferably used as an agent for the prophylaxis or treatment of Alzheimer's disease, frontotemporal lobar degeneration [progressive supranuclear palsy, corticobasal degeneration, frontotemporal dementia and parkinsonism linked to

MAPT mutation (FTDP-17), frontotemporal dementia, Pick's disease, argyrophilic grain dementia etc.], Charcot-Marie-Tooth disease and the like, particularly Alzheimer's disease or progressive supranuclear palsy.

(152) Here, the above-mentioned "prophylaxis" of a disease means, for example, administration of a medicament containing the compound of the present invention to patients who are expected to have a high risk of the onset due to some factor relating to the disease but have not developed the disease or patients who have developed the disease but do not have a subjective symptom, or administration of a medicament containing the compound of the present invention to patients who are feared to show recurrence of the disease after treatment of the disease.

(153) A medicament containing the compound of the present invention may be safely administered solely or by mixing with a pharmacologically acceptable carrier according to a method known per se (e.g., the method described in the Japanese Pharmacopoeia etc.) as the production method of a pharmaceutical preparation, and in the form of, for example, tablet (including sugar-coated tablet, film-coated tablet, sublingual tablet, orally disintegrating tablet, buccal etc.), pill, powder, granule, capsule (including soft capsule, microcapsule), troche, syrup, liquid, emulsion, suspension, release control preparation (e.g., immediate-release preparation, sustained-release preparation, sustained-release microcapsule), aerosol, film (e.g., orally disintegrating film, oral mucosa-adhesive film), injection (e.g., subcutaneous injection, intravenous injection, intramuscular injection, intraperitoneal injection), drip infusion, transdermal absorption type preparation, cream, ointment, lotion, adhesive preparation, suppository (e.g., rectal suppository, vaginal suppository), pellet, nasal preparation, pulmonary preparation (inhalant), eye drop and the like, orally or parenterally (e.g., intravenous, intramuscular, subcutaneous, intraorgan, intranasal, intradermal, instillation, intracerebral, intrarectal, intravaginal, intraperitoneal and intratumor administrations, administration to the vicinity of tumor and administration to the lesion).

(154) The content of the compound of the present invention in the medicament of the present invention is about 0.01 to 100% by weight of the entire medicament. The dose of the medicament of the present invention may vary depending on administration subject, administration route, disease and the like. For example, for oral administration to patients (body weight about 60 kg) with neurodegenerative disease (for example Alzheimer's disease, progressive supranuclear palsy, etc.), about 0.01 mg/kg body weight-about 50 mg/kg body weight, preferably about 0.05 mg/kg body weight-about 25 mg/kg body weight, more preferably about 0.1 mg/kg body weight-about 2 mg/kg body weight of an active ingredient (compound (I)) may be administered once to several portions per day.

(155) Examples of the pharmaceutically acceptable carrier include various organic or inorganic carrier materials that are conventionally used as preparation materials, for example, excipient, lubricant, binding agent and disintegrant for solid preparations; or solvent, solubilizing agent, suspending agent, isotonic agent, buffering agent, soothing agent and the like for liquid preparations. Furthermore, when necessary, ordinary additives such as preservative, antioxidant, colorant, sweetening agent, adsorbing agent, wetting agent and the like can be also used as appropriate in an appropriate amount.

(156) The dose of the medicament of the present invention as a sustained-release preparation varies depending on the kind and content of compound (I), dosage form, period of sustained drug release, subject animal of administration (e.g., mammals such as mouse, rat, hamster, guinea pig, rabbit, cat, dog, bovine, horse, swine, sheep, monkey, human etc.), and administration object. For example, for application by parenteral administration, about 0.1 to about 100 mg of compound (I) needs to be released from the administered preparation per 1 week.

(157) Examples of the excipient include lactose, white sugar, D-mannitol, starch, corn starch, crystalline cellulose, and light anhydrous silicic acid.

(158) Examples of the lubricant include magnesium stearate, calcium stearate, talc, and colloidal silica.

(159) Examples of the binding agent include crystalline cellulose, white sugar, D-mannitol, dextrin, hydroxypropylcellulose, hydroxypropylmethylcellulose, polyvinylpyrrolidone, starch, sucrose, gelatin, methylcellulose, and carboxymethylcellulose sodium.

(160) Examples of the disintegrant include starch, carboxymethylcellulose, carboxymethylcellulose calcium, carboxymethylstarch sodium, and L-hydroxypropylcellulose.

(161) Examples of the solvent include water for injection, alcohol, propylene glycol, Macrogol, sesame oil, corn oil, and olive oil.

(162) Examples of the solubilizing agent include polyethylene glycol, propylene glycol, D-mannitol, benzyl benzoate, ethanol, trisaminomethane, cholesterol, triethanolamine, and sodium carbonate, sodium citrate.

(163) Examples of the suspending agent include surfactants such as stearyl triethanolamine, sodium lauryl sulfate, laurylaminopropionic acid, lecithin, benzalkonium chloride, benzetonium chloride, glycerin monostearate and the like; and hydrophilic polymers such as polyvinyl alcohol, polyvinylpyrrolidone, carboxymethylcellulose sodium, methylcellulose, hydroxymethylcellulose, hydroxyethylcellulose, hydroxypropylcellulose and the like.

(164) Examples of the isotonic agent include glucose, D-sorbitol, sodium chloride, glycerin, and D-mannitol.

(165) Examples of the buffering agent include buffer solutions such as phosphates, acetates, carbonates, and citrates.

(166) Examples of the soothing agent include benzyl alcohol.

(167) Examples of the preservative include parahydroxybenzoates, chlorobutanol, benzyl alcohol, phenylethyl alcohol, dehydroacetic acid, and sorbic acid.

(168) Examples of the antioxidant include sulfites, ascorbic acid, and α -tocopherol.

(169) For the prophylaxis or treatment of various diseases, the compound of the present invention may also be used together with other drug (hereinafter, to be referred to as concomitant drug). In the following, a medicament to be used when the compound of the present invention is used together with other drug is referred to as “the combination agent of the present invention”.

(170) For example, when the compound of the present invention is used as a HDAC inhibitor, preferably a class II HDAC inhibitor, more preferably a HDAC6 inhibitor, it may be used together with the following drugs. tranquilizer (diazepam, lorazepam, clorazepate dipotassium, oxazepam, chlordiazepoxide, medazepam, oxazolam, cloxazolam, clotiazepam, bromazepam, etizolam, fludiazepam, hydroxyzine, nitrazepam, triazolam, alprazolam etc.), antipsychotic (chlorpromazine hydrochloride, prochlorperazine, trifluoperazine, thioridazine hydrochloride, perphenazine maleate, fluphenazine enanthate, prochlorperazine maleate, levomepromazine maleate, promethazine hydrochloride, haloperidol, clozapine, trifluoperazine dihydrochloride, fluphenazine hydrochloride, olanzapine, quetiapine fumarate, risperidone, aripiprazole, bromperidol, spiperone, reserpine, clocapramine hydrochloride, sulpiride, zotepine, tiotixene etc.), antiepileptic drug (phenytoin, ethosuximide, acetazolamide, chlordiazepoxide, trimethadione, carbamazepine, phenobarbital, primidone, sulthiame, sodium valproate, clonazepam, diazepam, nitrazepam etc.), antidepressant and therapeutic drug for manic psychosis [tricyclic or tetracyclic antidepressant drug (imipramine hydrochloride, clomipramine hydrochloride, desipramine hydrochloride, amitriptyline hydrochloride, nortriptyline hydrochloride, amoxapine, mianserin hydrochloride, maprotiline hydrochloride, etc.), noxiptiline, phenelzine, sulpiride, trazodone hydrochloride, lithium carbonate, selective serotonin reuptake inhibitor (fluvoxamine maleate, fluoxetine hydrochloride, citalopram hydrobromide, sertraline hydrochloride, paroxetine hydrochloride, paroxetine hydrochloride hydrate, escitalopram oxalate etc.), serotonin-noradrenalin reuptake inhibitor (venlafaxine hydrochloride, duloxetine hydrochloride, venlafaxine hydrochloride etc.), noradrenalin reuptake inhibitor (reboxetine mesylate etc.), noradrenalin-dopamine reuptake inhibitor (bupropion hydrochloride etc.), mirtazapine, trazodone hydrochloride, nefazodone hydrochloride, setiptiline maleate, 5-HT_{1A} agonist (buspirone hydrochloride, tandospirone citrate, osetozotan

hydrochloride) etc.] benzodiazepine (clonazepam etc.), L-type calcium channel inhibitor (pregabalin etc.), 5-HT_{1A} agonist (buspirone hydrochloride, tandospirone citrate, osetozotan hydrochloride etc.), 5-HT₃ antagonist (cyamemazine etc.), heart non-selective β inhibitor (propranolol hydrochloride, oxiprenolol hydrochloride etc.), histamine H₁ antagonist (hydroxyzine hydrochloride etc.), therapeutic drug for schizophrenia (chlorpromazine, haloperidol, sulpiride, clozapine, trifluoperazine hydrochloride, fluphenazine hydrochloride, olanzapine, quetiapine fumarate, risperidone, aripiprazole etc.), CRF antagonist, other antianxiety drug (meprobamate etc.), tachykinin antagonist (MK-869, saredutant etc.), drug that acts on metabotropic glutamate receptor, CCK antagonist, B3 adrenaline antagonist (amibegron hydrochloride etc.), GAT-1 inhibitor (tiagabine hydrochloride etc.), N-type calcium channel inhibitor, type II carbonic anhydrase inhibitor, NMDA glycine site agonist, NMDA antagonist (memantine etc.), peripheral benzodiazepine receptor agonist, vasopressin antagonist, vasopressin V1b antagonist, vasopressin V1a antagonist, phosphodiesterase inhibitor, opioids antagonist, opioids agonist, uridine, nicotinic acid receptor agonists, thyroid hormone (T3, T4), TSH, TRH, MAO inhibitor (phenelzine sulfate, tranlycypromine sulfate, moclobemide etc.), 5-HT_{2A} antagonist, 5-HT_{2A} inverse agonist, COMT inhibitor (entacapone, etc.), therapeutic drug for bipolar disorder (lithium carbonate, sodium valproate, lamotrigine, riluzole, felbamate etc.), cannabinoid CB1 antagonist (rimonabant etc.), FAAH inhibitor, sodium channel inhibitor, anti-ADHD drug (methylphenidate hydrochloride, methamphetamine hydrochloride etc.), therapeutic drug for alcoholism, therapeutic drug for autism, therapeutic drug for chronic fatigue syndrome, therapeutic drug for convulsion, therapeutic drug for fibromyalgia, therapeutic drug for headache, therapeutic drug for insomnia (etizolam, zopiclone, triazolam, xolpidem, ramelteon, indiplon etc.), therapeutic drug for quitting smoking, therapeutic drug for myasthenia gravis, therapeutic drug for cerebral infarction, therapeutic drug for manic psychosis, therapeutic drug for hypersomnia, therapeutic drug for pain, therapeutic drug for dysthymia, therapeutic drug for dysautonomia, therapeutic drug for male and female sexual dysfunction, therapeutic drug for migraine, therapeutic drug for pathological gambling, therapeutic drug for restless legs syndrome, therapeutic drug for substance dependence, therapeutic drug for alcohol-related disease, therapeutic drug for irritable bowel syndrome, therapeutic drug for Alzheimer's disease (donepezil, galanthamine, memantine, rivastigmine, etc.), therapeutic drug for Parkinson's disease (levodopa, carbidopa, benserazide, selegiline, rasagiline, zonisamide, entacapone, amantadine, talipexole, pramipexole, ropinirole, rotigotine, apomorphine, cabergoline, pergolide, bromocriptine, istradefylline, trihexyphenidyl, biperiden, piroheptine, profenamine, promethazine, droxidopa, amantadine hydrochloride, bromocriptine mesilate, trihexyphenidyl hydrochloride, selegiline hydrochloride, combination thereof etc.), therapeutic drug for Parkinson's disease associated with dementia (rivastigmine), therapeutic drug for Lewy body dementia (donepezil), therapeutic drug for ALS (riluzole, neurotrophic factor, etc.), therapeutic drug for hyperlipidemia such as cholesterol-lowering drug (statin series (pravastatin sodium, atorvastatin, simvastatin, rosuvastatin, etc.), fibrate (clofibrate etc.), squalene synthase inhibitor), therapeutic drug for abnormal behavior or dementia-related wandering (sedative drug, antianxiety drug, etc.), apoptosis inhibitor, antiobesity drug, antidiabetic drug, therapeutic drug for hypertension, therapeutic drug for hypotension, therapeutic drug for rheumatism (DMARD), anticancer drug, therapeutic drug for hypoparathyroidism (PTH), calcium receptor antagonist, sex hormone or derivative thereof (progesterone, estradiol, estradiol benzoate etc.), neuron differentiation accelerator, neurogeneration promotor, non-steroidal anti-inflammatory drug (meloxicam, tenoxicam, indomethacin, ibuprofen, celecoxib, rofecoxib, aspirin etc.), steroid (dexamethasone, cortisone acetate, etc.), anti-cytokine drug (TNF inhibitor, MAP kinase inhibitor, etc.), antibody medicament, nucleic acid or nucleic acid derivative, aptamer and the like.

(171) For combined use, the administration time of the compound of the present invention and the concomitant drug is not restricted, and the compound of the present invention or the concomitant drug can be administered to an administration subject simultaneously, or may be administered at

different times. The dosage of the concomitant drug may be determined according to the dose clinically used, and can be appropriately selected depending on an administration subject, administration route, disease, combination and the like.

(172) The administration form of the combined use is not particularly limited, and the compound of the present invention and a concomitant drug only need to be combined on administration. Examples of such administration mode include the following: (1) administration of a single preparation obtained by simultaneously processing the compound of the present invention and the concomitant drug, (2) simultaneous administration of two kinds of preparations of the compound of the present invention and the concomitant drug, which have been separately produced, by the same administration route, (3) administration of two kinds of preparations of the compound of the present invention and the concomitant drug, which have been separately produced, by the same administration route in a staggered manner, (4) simultaneous administration of two kinds of preparations of the compound of the present invention and the concomitant drug, which have been separately produced, by different administration routes, (5) administration of two kinds of preparations of the compound of the present invention and the concomitant drug, which have been separately produced, by different administration routes in a staggered manner (e.g., administration in the order of the compound of the present invention and the concomitant drug, or in the reverse order) and the like.

(173) The mixing ratio of the compound of the present invention and a concomitant drug in the combination agent of the present invention can be appropriately selected based on the subject of administration, administration route, disease and the like.

(174) For example, while the content of the compound of the present invention in the combination agent of the present invention varies depending on the preparation form, it is generally about 0.01-100 wt %, preferably about 0.1-50 wt %, more preferably about 0.5-20 wt %, of the whole preparation.

(175) The content of the concomitant drug in the combination agent of the present invention varies depending on the preparation form, and generally about 0.01 to 100% by weight, preferably about 0.1 to 50% by weight, further preferably about 0.5 to 20% by weight, of the entire preparation.

(176) While the content of the additive such as a carrier and the like in the combination agent of the present invention varies depending on the form of a preparation, it is generally about 1 to 99.99% by weight, preferably about 10 to 90% by weight, based on the preparation.

(177) When the compound of the present invention and the concomitant drug are separately prepared, the same content may be adopted.

(178) Any amount of the concomitant drug can be adopted as long as the side effects do not cause a problem. The daily dosage in terms of the concomitant drug varies depending on the severity, age, sex, body weight, sensitivity difference of the subject, administration period, interval, and nature, pharmacology, kind of the pharmaceutical preparation, kind of effective ingredient, and the like, and not particularly restricted, and the amount of a drug is, in the case of oral administration for example, generally about 0.001 to 2000 mg, preferably about 0.01 to 500 mg, further preferably about 0.1 to 100 mg, per 1 kg of a mammal and this is generally administered once to 4-times, divided in a day.

(179) When the combination agent of the present invention is administered, the compound of the present invention and the concomitant drug can be administered simultaneously, or may be administered in a staggered manner. When administered at a time interval, the interval varies depending on the effective ingredient, dosage form and administration method, and, for example, when the concomitant drug is administered first, a method in which the compound of the present invention is administered within time range of from 1 minute to 3 days, preferably from 10 minutes to 1 day, more preferably from 15 minutes to 1 hour, after administration of the concomitant drug is an example. When the compound of the present invention is administered first, a method in which the concomitant drug is administered within time range of from 1 minute to 1 day, preferably from

10 minutes to 6 hours, more preferably from 15 minutes to 1 hour after administration of the compound of the present invention is an example.

EXAMPLES

(180) The present invention is explained in detail in the following by referring to Examples, Experimental Examples and Formulation Examples, which are not to be construed as limitative, and the invention may be changed within the scope of the present invention.

(181) In the following Examples, the “room temperature” generally means about 10° C. to about 35° C. The ratios indicated for mixed solvents are volume mixing ratios, unless otherwise specified. % means wt %, unless otherwise specified.

(182) The elution by column chromatography in the Examples was performed under the observation by TLC (Thin Layer Chromatography) unless otherwise specified. In the observation by TLC, 60 F.sub.254 manufactured by Merck was used as a TLC plate, the solvent used as an elution solvent in column chromatography was used as a developing solvent, and UV detector was used for the detection. In silica gel column chromatography, the indication of NH means use of aminopropylsilane-bonded silica gel, and the indication of Diol means use of 3-(2,3-dihydroxypropoxy)propylsilane-bonded silica gel. In preparative HPLC (high performance liquid chromatography), the indication of C18 means use of octadecyl-bonded silica gel. The ratio for elution solvent is, unless otherwise specified, a volume mixing ratio.

(183) For the analysis of ¹H NMR, ACD/SpecManager (trade name) software and the like were used. Peaks of a hydroxyl group, an amino group and the like, having very mild proton peak, are not sometimes described.

(184) MS (mass spectrum) was measured by LC/MS (liquid chromatograph mass spectrometer). As the ionization method, ESI (Electrospray Ionization) method, or APCI (Atmospheric Pressure Chemical Ionization) method was used. The data indicates actual measured value (found). While molecular ion peak is generally observed, a fragment ion is sometimes observed. In the case of a salt, a molecular ion peak or fragment ion peak of free form is generally observed. Elemental analysis value (Anal.) is described as calculated value (Calcd) and actual measured value (Found).

(185) Powder X-RAY diffraction pattern was measured using Cu-K α characteristic radiation from Rigaku Ultima IV, and characteristic peaks were described.

(186) In the following Examples, the following abbreviations are used. MS: mass spectrum M: mol concentration CDCl₃: deuteriochloroform DMSO-d₆: deuterodimethyl sulfoxide ¹H NMR: proton nuclear magnetic resonance LC/MS: liquid chromatograph mass spectrometer ESI: Electrospray Ionization APCI: Atmospheric Pressure Chemical Ionization AIBN: 2,2'-azobis(isobutyronitrile) Boc: tert-butoxycarbonyl CDI: 1,1'-carbonyldiimidazole CO₂: carbon dioxide DIPEA: N,N-diisopropylethylamine DMF: N,N-dimethylformamide DMSO: dimethyl sulfoxide NBS: N-bromosuccinimide TEA: triethylamine THF: tetrahydrofuran XANTPHOS: 4,5-bis(diphenylphosphino)-9,9-dimethylxanthene

Example 1 and Example 2

6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 1)

6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 2)

A) tert-butyl 2-[3-(methoxycarbonyl)-4-methylbenzoyl]hydrazine-1-carboxylate

(187) The following reaction was carried out by dividing the described scale into 5 reactions. A mixture of methyl 5-bromo-2-methylbenzoate (42.0 g), tert-butyl hydrazinecarboxylate (29.1 g), bis(dibenzylideneacetone)palladium (0) (5.27 g), XANTPHOS (5.30 g), N,N-dicyclohexylmethylamine (58.3 mL) and cyclopentyl methyl ether (1000 mL) was stirred under carbon monoxide atmosphere (0.5 MPa) at 95° C. for 5 hr. The five reaction mixtures were combined, and filtered through Celite, and the filtrate was concentrated under reduced pressure. The obtained residue was dissolved in a mixture of ethyl acetate and THF, and the solution was

washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (NH, ethyl acetate), and crystallized from hexane/THF to give the title compound (45.7 g). ¹H NMR (300 MHz, DMSO-d₆) δ 1.32-1.53 (9H, m), 2.57 (3H, s), 3.86 (3H, s), 7.46 (1H, d, J=7.9 Hz), 7.94 (1H, dd, J=7.9, 1.5 Hz), 8.33 (1H, s), 8.93 (1H, s), 10.30 (1H, s).

(188) MS: [M-H].sup.- 307.0.

B) methyl 5-(hydrazinecarbonyl)-2-methylbenzoate hydrochloride

(189) To tert-butyl 2-[3-(methoxycarbonyl)-4-methylbenzoyl]hydrazine-1-carboxylate (45.7 g) was added 4 M hydrogen chloride cyclopentyl methyl ether solution (200 mL) at room temperature.

The mixture was stirred at room temperature for 3 days. The precipitated solid was collected by filtration, and dried to give the title compound (36.3 g). ¹H NMR (300 MHz, DMSO-d₆) δ 2.59 (3H, s), 3.88 (3H, s), 7.52 (1H, d, J=8.1 Hz), 8.03 (1H, dd, J=7.9, 2.1 Hz), 8.36 (1H, d, J=1.9 Hz), 9.71-10.80 (2H, m), 11.64 (1H, br s).

(190) MS: [M+H].sup.+ 209.0.

C) methyl 5-(5-(difluoromethyl)-1,3,4-oxadiazol-2-yl)-2-methylbenzoate

(191) To a mixture of methyl 5-(hydrazinecarbonyl)-2-methylbenzoate hydrochloride (36.3 g) and THF (750 mL) was added DIPEA (129 mL) at room temperature. To the mixture was added dropwise difluoroacetic acid anhydride (27.6 mL) at 0° C. The mixture was stirred at room temperature for 2 hr, and to the mixture was added 4-methylbenzenesulfonyl chloride (56.5 g) at room temperature. The mixture was stirred overnight at room temperature, to the mixture was added water at room temperature, and the mixture was extracted with ethyl acetate.

(192) The organic layer was separated, washed with saturated aqueous sodium hydrogencarbonate solution, water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was crystallized from hexane/ethyl acetate to give the title compound (24.9 g).

(193) .sup.1H NMR (300 MHz, DMSO-d₆) δ 2.63 (3H, s), 3.90 (3H, s), 7.37-7.75 (2H, m), 8.15 (1H, dd, J=8.1, 2.1 Hz), 8.44 (1H, d, J=2.3 Hz).

D) methyl 2-(bromomethyl)-5-((5-(difluoromethyl)-1,3,4-oxadiazol-2-yl))benzoate

(194) To a mixture of methyl 5-(5-(difluoromethyl)-1,3,4-oxadiazol-2-yl)-2-methylbenzoate (14.0 g) and benzotrifluoride (500 mL) were added NBS (16.7 g) and AIBN (0.857 g) at room temperature. The mixture was stirred under argon atmosphere at 90° C. for 1 hr. To the mixture was added water at room temperature, and the mixture was extracted with ethyl acetate. The organic layer was separated, washed with saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. To the mixture of the residue and THF (500 mL) was added DIPEA (10.0 mL) at room temperature. To the mixture was added dropwise diethyl phosphonate (7.40 mL) at 0° C., and the mixture was stirred overnight at room temperature. To the mixture was added water at room temperature, and the mixture was extracted with ethyl acetate. The organic layer was separated, washed with saturated aqueous ammonium chloride solution, water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (ethyl acetate/hexane), and crystallized from hexane/ethyl acetate to give the title compound (12.7 g).

(195) .sup.1H NMR (300 MHz, CDCl₃) δ 4.01 (3H, s), 5.01 (2H, s), 6.94 (1H, t, J=51.6 Hz), 7.69 (1H, d, J=8.3 Hz), 8.24 (1H, dd, J=7.9, 1.9 Hz), 8.71 (1H, d, J=1.9 Hz).

(196) MS: [M+H].sup.+ 347.0.

E) tert-butyl [(4-methylbenzene-1-sulfonyl) (pyridin-2-yl)methyl]carbamate

(197) To a mixture of sodium p-toluenesulfinate (25.0 g), tert-butyl carbamate (10.9 g) and water (140 mL) was added a mixture of pyridine-2-carbaldehyde (10.0 g) and methanol (70 mL) at room temperature. The mixture was stirred at room temperature for 30 min until the suspension became a clear solution. To the mixture was added formic acid (7.16 mL) at room temperature, and the mixture was stirred at room temperature for 3 days. The precipitated solid was collected by

filtration, and washed with water and diisopropyl ether to give the title compound (23.0 g).

(198) .sup.1H NMR (300 MHz, DMSO-d.sub.6) δ 1.02-1.31 (9H, m), 2.39 (3H, s), 6.06 (1H, br d, $J=10.2$ Hz), 7.31-7.52 (3H, m), 7.66 (2H, d, $J=8.3$ Hz), 7.77-7.84 (1H, m), 7.85-8.00 (1H, m), 8.15 (1H, br d, $J=10.5$ Hz), 8.49-8.63 (1H, m).

F) tert-butyl [2-oxo-1,2-di(pyridin-2-yl)ethyl]carbamate

(199) A mixture of tert-butyl [(4-methylbenzene-1-sulfonyl) (pyridin-2-yl)methyl]carbamate (11.0 g), pyridine-2-carbaldehyde (3.41 g), 3-ethyl-5-(2-hydroxyethyl)-4-methylthiazolium bromide (1.53 g) and THF (121 mL) was degassed, and to the mixture was added dropwise triethylamine (63.3 mL) under argon atmosphere at room temperature. The mixture was again degassed, and the mixture was stirred overnight under argon atmosphere at 60° C. To the mixture was added saturated aqueous ammonium chloride solution at room temperature, and the mixture was partitioned between water and ethyl acetate. The separated organic layer was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (NH, ethyl acetate/hexane) to give the title compound (9.81 g) as a crude product. The obtained compound was used in the next step without further purification.

(200) MS: [M+H-Boc].sup.+214.0.

G) tert-butyl [(1RS,2RS)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]carbamate

(201) To a mixture of tert-butyl [2-oxo-1,2-di(pyridin-2-yl)ethyl]carbamate (9.81 g), methanol (60 mL) and THF (60 mL) was added sodium borohydride (1.72 g) at 0° C., and the mixture was stirred overnight at room temperature. To the mixture was added water at 0° C., and the mixture was stirred at room temperature for 1 hr, and partitioned between water and ethyl acetate. The separated organic layer was washed with water and saturated brine. The combined aqueous layer was re-extracted with ethyl acetate, and the extract was washed with water and saturated brine. The combined organic layer was dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was solidified from ethyl acetate/hexane, and the obtained solid was recrystallized from ethyl acetate/hexane to give the title compound (6.64 g).

(202) .sup.1H NMR (300 MHz, DMSO-d.sub.6) δ 1.11-1.35 (9H, m), 4.82-5.09 (2H, m), 5.61 (1H, d, $J=5.1$ Hz), 7.02 (1H, br d, $J=8.7$ Hz), 7.10 (1H, br d, $J=7.9$ Hz), 7.15-7.30 (3H, m), 7.57-7.80 (2H, m), 8.42 (1H, d, $J=4.3$ Hz), 8.52 (1H, br d, $J=4.1$ Hz).

(203) MS: [M+H].sup.+ 316.1.

H) 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoinol-1-one

(204) To a mixture of tert-butyl [(1RS,2RS)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]carbamate (2.21 g) and acetic acid (20 mL) was added dropwise hydrogen bromide (30% acetic acid solution (ca. 5.1 M), 11.0 mL) at room temperature, and the mixture was stirred overnight at room temperature. The mixture was concentrated under reduced pressure, and subjected to azeotropy with toluene and toluene/ethanol to give a white solid (3.54 g). To a mixture of the obtained solid (2.83 g), methyl 2-(bromomethyl)-5-(5-(difluoromethyl)-1,3,4-oxadiazol-2-yl)benzoate (2.15 g) and DMF (50 mL) was added dropwise DIPEA (6.46 mL) at 0° C., and the mixture was stirred at room temperature for 3 hr. To the mixture was added dropwise acetic acid (2.12 mL) at room temperature, and the mixture was stirred overnight at room temperature. The mixture was poured into water at room temperature, and extracted with ethyl acetate. The extract was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (methanol/ethyl acetate), and crystallized from ethyl acetate/diisopropyl ether to give the title compound (2.36 g).

(205) .sup.1H NMR (300 MHz, DMSO-d.sub.6) δ 4.75 (1H, d, $J=18.8$ Hz), 5.03 (1H, d, $J=18.8$ Hz), 5.49 (1H, dd, $J=8.8, 5.8$ Hz), 5.76 (1H, d, $J=9.0$ Hz), 5.98 (1H, d, $J=5.6$ Hz), 7.17 (1H, ddd, $J=7.3, 4.9, 1.3$ Hz), 7.30-7.72 (4H, m), 7.72-7.89 (3H, m), 8.04 (1H, d, $J=0.8$ Hz), 8.17-8.24 (1H, m), 8.25-8.31 (1H, m), 8.60-8.66 (1H, m).

(206) MS: [M+H].sup.+ 450.1.

I) 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 1) and

6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 2)

(207) 6-[5-(Difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one was optically resolved by preparative SFC (column: CHIRALCEL OD-H, 20 mmID×250 mmL, 5 µm, mobile phase: CO.sub.2/ethanol=770/230 (v/v)).

(208) The fraction of the optical isomer having a longer retention time was concentrated under reduced pressure, and the residue was crystallized from hexane/ethyl acetate to give a white solid (1.20 g). The obtained solid (1.17 g) was dissolved in ethyl acetate (12 mL) at 70° C., and to the mixture was added dropwise heptane (10 mL) at 70° C. The mixture was stirred at 70° C. for 30 min, and then to the mixture was added dropwise additional heptane (30 mL) at 50° C.-75° C. The mixture was stirred at 50° C. for 30 min, and then at room temperature for 1 hr. The precipitated solid was collected by filtration, and washed with heptane to give 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (1.09 g, >99% e.e., retention time: 5.48 min (analysis column, column: CHIRALCEL OD-H, 4.6 mmID×150 mmL, 5 µm, mobile phase: CO.sub.2/ethanol=770/230 (v/v))). The absolute stereochemistry of the compound was determined by X-ray crystallography.

(209) .sup.1H NMR (300 MHz, DMSO-d.sub.6) δ 4.75 (1H, d, J=18.8 Hz), 5.03 (1H, d, J=18.8 Hz), 5.49 (1H, dd, J=8.8, 5.5 Hz), 5.76 (1H, d, J=9.0 Hz), 5.98 (1H, d, J=5.6 Hz), 7.17 (1H, ddd, J=7.3, 4.9, 1.3 Hz), 7.28-7.72 (4H, m), 7.72-7.91 (3H, m), 8.04 (1H, d, J=0.8 Hz), 8.18-8.24 (1H, m), 8.25-8.32 (1H, m), 8.60-8.67 (1H, m).

(210) MS: [M+H].sup.+ 450.1.

(211) Anal. Calcd for C.sub.23H.sub.17F.sub.2N.sub.5O.sub.3: C, 61.47; H, 3.81; N, 15.58. Found: C, 61.68; H, 4.01; N, 15.80.

(212) powder X-RAY crystal diffraction pattern (diffraction angle 2θ): 5.0°, 7.1°, 9.1°, 10.8°, 16.6°, 18.3°, 19.8°, 21.6°, 22.8°, 23.7°, 26.1°

(213) The fraction of the optical isomer having a shorter retention time was concentrated under reduced pressure, and the residue was dissolved in ethyl acetate (15 mL) at 70° C. To the mixture was added dropwise diisopropyl ether (30 mL) at 50° C.-70° C., and the obtained suspension was stirred at 50° C. for 1 hr. To the mixture was added dropwise diisopropyl ether (30 mL) at 50° C., and the mixture was stirred at 50° C. for 30 min, and then at room temperature for 1 hr. The precipitated solid was collected by filtration, and washed with diisopropyl ether to give 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (1.02 g, >99% e.e., retention time: 4.52 min (analysis column: column: CHIRALCEL OD-H, 4.6 mmID×150 mmL, 5 µm, mobile phase: CO.sub.2/ethanol=770/230 (v/v))).

(214) MS: [M+H].sup.+ 450.1.

Example 3 and Example 4

6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 3)

6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 4)

A) tert-butyl [2-(5-fluoropyridin-2-yl)-2-oxo-1-(pyridin-2-yl)ethyl]carbamate

(215) A mixture of tert-butyl [(4-methylbenzene-1-sulfonyl) (pyridin-2-yl)methyl]carbamate (12.1 g), 5-fluoropyridine-2-carbaldehyde (4.18 g), 3-ethyl-5-(2-hydroxyethyl)-4-methylthiazolium bromide (1.68 g) and THF (134 mL) was degassed, and to the mixture was added dropwise triethylamine (69.6 mL) under argon atmosphere at room temperature. The mixture was again degassed, and the mixture was stirred overnight under argon atmosphere at 60° C. To the mixture

was added saturated aqueous ammonium chloride solution at 0° C., and the mixture was partitioned between water and ethyl acetate. The separated organic layer was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (NH, ethyl acetate/hexane), and crystallized from ethyl acetate/hexane to give the title compound (7.14 g).

(216) .sup.1H NMR (300 MHz, DMSO-d.sub.6) δ 1.12-1.51 (9H, m), 6.76 (1H, d, J=8.7 Hz), 7.25 (1H, ddd, J=7.5, 4.9, 1.1 Hz), 7.45 (1H, br d, J=8.3 Hz), 7.53 (1H, d, J=7.9 Hz), 7.78 (1H, td, J=7.7, 1.9 Hz), 7.90 (1H, td, J=8.7, 2.6 Hz), 8.11 (1H, dd, J=8.8, 4.7 Hz), 8.33 (1H, br d, J=4.5 Hz), 8.64 (1H, d, J=2.6 Hz).

B) tert-butyl [(1RS,2RS)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]carbamate (217) To a mixture of tert-butyl [2-(5-fluoropyridin-2-yl)-2-oxo-1-(pyridin-2-yl)ethyl]carbamate (6.59 g) and THF (100 mL) was added dropwise potassium tri(sec-butyl)borohydride (1 M THF solution, 23.9 mL) under argon atmosphere at -78° C., and the mixture was stirred at -78° C. for 2 hr. To the mixture was added water (40 mL) at -78° C., and to the mixture were added 2 M aqueous sodium hydroxide solution (99 mL) and 30% aqueous hydrogen peroxide (20.3 mL) while keeping the internal temperature 0° C. or below. The mixture was stirred at room temperature for 2 hr. Then, sodium thiosulfate pentahydrate (49.4 g) was dissolved in water (190 mL), and the obtained solution was added dropwise to the mixture at 0° C. The mixture was stirred at room temperature for 2 hr, and extracted with ethyl acetate. The extract was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was recrystallized from ethyl acetate/heptane to give the title compound (5.62 g).

(218) .sup.1H NMR (300 MHz, DMSO-d.sub.6) δ 1.02-1.46 (9H, m), 4.84-5.10 (2H, m), 5.69 (1H, br d, J=4.5 Hz), 6.98 (1H, br d, J=8.7 Hz), 7.13-7.25 (2H, m), 7.29 (1H, dd, J=8.7, 4.5 Hz), 7.50-7.78 (2H, m), 8.43 (1H, dd, J=4.9, 0.8 Hz), 8.51 (1H, d, J=2.6 Hz).

(219) MS: [M+H].sup.+234.0.

C) 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one

(220) To a mixture of tert-butyl [(1RS,2RS)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]carbamate (4.09 g) and acetic acid (40 mL) was added hydrogen bromide (25% acetic acid solution, 10 mL) at room temperature. To the mixture was added acetic acid (20 mL) at room temperature, and the mixture was stirred at room temperature for 30 min. The mixture was concentrated under reduced pressure, and the residue was subjected three times to azeotropy with toluene, and dried. To a mixture of the obtained residue, DIPEA (13.5 mL) and DMF (29 mL) was added dropwise a solution prepared by dissolving methyl 2-(bromomethyl)-5-(5-(difluoromethyl)-1,3,4-oxadiazol-2-yl)benzoate (3.87 g) in DMF (10 mL), at room temperature, and the mixture was stirred at room temperature for 60 hr. To the mixture was added water, and the mixture was extracted with ethyl acetate. The separated organic layer was washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (ethyl acetate/hexane). The obtained solid was suspended in diisopropyl ether/ethyl acetate, and collected by filtration. The solid was suspended in hexane/ethyl acetate=1/1 (50 mL), sonicated, and stirred at 50° C. for 10 min. The solid was collected by filtration, and washed with hexane/ethyl acetate to give the title compound (4.09 g).

(221) .sup.1H NMR (300 MHz, DMSO-d.sub.6) δ 4.70 (1H, d, J=19.6 Hz), 4.97 (1H, d, J=20.0 Hz), 5.47-5.57 (1H, m), 5.72 (1H, d, J=9.0 Hz), 6.05 (1H, d, J=5.3 Hz), 7.31-7.74 (5H, m), 7.77-7.87 (2H, m), 8.05 (1H, d, J=1.1 Hz), 8.22 (1H, dd, J=7.9, 1.5 Hz), 8.26 (1H, t, J=1.5 Hz), 8.57-8.69 (1H, m).

(222) MS: [M+H].sup.+ 468.1.

D) 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 3) and

6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 4)

(223) 6-[5-(Difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (3.09 g) was optically resolved by preparative HPLC (column: CHIRALCEL OD, 50 mmID×500 mmL, 20 μm, mobile phase: hexane/ethanol=700/300 (v/v)).

(224) The fraction of the optical isomer having a longer retention time was concentrated under reduced pressure, and the residue was crystallized from heptane/isopropyl acetate to give 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (1.35 g, 99% e.e., retention time: 9.16 min (analysis condition, column: CHIRALCEL OD-H, 4.6 mmID×250 mmL, 5 μm, mobile phase: hexane/ethanol=700/300 (v/v)). The absolute stereochemistry of the compound was determined by X-ray crystallography.

(225) ¹H NMR (300 MHz, DMSO-d₆) δ 4.71 (1H, d, J=18.8 Hz), 4.97 (1H, d, J=18.8 Hz), 5.52 (1H, dd, J=9.0, 5.3 Hz), 5.72 (1H, d, J=9.0 Hz), 6.05 (1H, d, J=5.3 Hz), 7.31-7.74 (5H, m), 7.77-7.89 (2H, m), 8.05 (1H, d, J=1.1 Hz), 8.22 (1H, dd, J=8.1, 1.7 Hz), 8.26 (1H, t, J=1.5 Hz), 8.53-8.72 (1H, m).

(226) MS: [M+H]⁺ 468.1.

(227) Anal. Calcd for C₂₃H₁₆F₃N₅O₃·3H₂O·C, 58.43; H, 3.54; N, 14.81. Found: C, 58.53; H, 3.53; N, 14.78.

(228) powder X-RAY crystal diffraction pattern (diffraction angle 2θ): 7.2°, 11.3°, 16.2°, 19.2°, 22.1°, 22.7°, 24.2°, 26.5°

(229) The fraction of the optical isomer having a shorter retention time was concentrated under reduced pressure, and the residue was crystallized from heptane/isopropyl acetate to give 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (1.37 g, 99% e.e., retention time: 7.35 min (analysis condition, column: CHIRALCEL OD-H, 4.6 mmID×250 mmL, 5 μm, mobile phase: hexane/ethanol=700/300 (v/v)).

(230) MS: [M+H]⁺ 468.1.

Example 5 and Example 6

2-[(1R,2R)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 5)

2-[(1S,2S)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 6)

A) tert-butyl [(5-fluoropyridin-2-yl) (4-methylbenzene-1-sulfonyl)methyl]carbamate

(231) To a mixture of sodium p-toluenesulfinate (21.4 g), tert-butyl carbamate (9.36 g) and water (120 mL) was added a solution prepared by dissolving 5-fluoropyridine-2-carbaldehyde (10 g) in methanol (60 mL), at room temperature. The mixture was stirred at room temperature for 30 min until the suspension became a clear solution. To the mixture was added formic acid (6.13 mL) at room temperature, and the mixture was stirred at room temperature for 3 days. The precipitated solid was collected by filtration, washed with water and diisopropyl ether, and dried under reduced pressure to give the title compound (24.4 g).

(232) ¹H NMR (300 MHz, DMSO-d₆) 1.10-1.28 (9H, m), 2.39 (3H, s), 6.09 (1H, d, J=10.5 Hz), 7.44 (2H, d, J=7.9 Hz), 7.67 (2H, d, J=7.9 Hz), 7.81-7.97 (2H, m), 8.26 (1H, d, J=10.2 Hz), 8.56 (1H, d, J=2.3 Hz).

B) tert-butyl [1,2-bis(5-fluoropyridin-2-yl)-2-oxoethyl]carbamate

(233) A mixture of tert-butyl [(5-fluoropyridin-2-yl) (4-methylbenzene-1-sulfonyl)methyl]carbamate (12.7 g), 5-fluoropyridine-2-carbaldehyde (4.18 g), 3-ethyl-5-(2-hydroxyethyl)-4-methylthiazolium bromide (1.68 g) and THF (134 mL) was degassed, and to the mixture was added triethylamine (69.6 mL) under argon atmosphere at room temperature. The

mixture was again degassed, and the mixture was stirred overnight under argon atmosphere at 60° C. To the mixture was added saturated aqueous ammonium chloride solution at 0° C., and the mixture was partitioned between water and ethyl acetate. The separated organic layer was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (NH, ethyl acetate/hexane), and crystallized from hexane/ethyl acetate to give the title compound (7.68 g).

(234) ¹H NMR (300 MHz, DMSO-d₆) δ 1.13-1.58 (9H, m), 6.74 (1H, d, J=8.7 Hz), 7.54 (1H, br d, J=8.7 Hz), 7.58-7.67 (1H, m), 7.68-7.81 (1H, m), 7.91 (1H, td, J=8.8, 2.8 Hz), 8.12 (1H, dd, J=8.7, 4.9 Hz), 8.34 (1H, d, J=2.6 Hz), 8.63 (1H, d, J=3.0 Hz).

(235) MS: [M+H-Boc].sup.+250.0.

C) tert-butyl [(1R,2R)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]carbamate

(236) To a mixture of tert-butyl [1,2-bis(5-fluoropyridin-2-yl)-2-oxoethyl]carbamate (7.0 g) and THF (100 mL) was added dropwise potassium tri(sec-butyl)borohydride (1 M THF solution, 30.1 mL) under argon atmosphere at -78° C., and the mixture was stirred at -78° C. for 2 hr. To the mixture was added water (20 mL) at -78° C., and 2 M aqueous sodium hydroxide solution (100 mL) and 30% aqueous hydrogen peroxide (20.5 mL) were added thereto while keeping the internal temperature 0° C. or below. The mixture was stirred at room temperature for 4 hr. Sodium thiosulfate pentahydrate (49.7 g) was dissolved in water (200 mL), the obtained solution was added to the mixture at 0° C., and the mixture was stirred at room temperature for 2 hr. The mixture was extracted with ethyl acetate. The extract was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (ethyl acetate/hexane), and crystallized from hexane/ethyl acetate to give the title compound (5.97 g).

(237) ¹H NMR (300 MHz, DMSO-d₆) δ 1.01-1.48 (9H, m), 4.84-4.95 (1H, m), 4.96-5.10 (1H, m), 5.67 (1H, d, J=5.6 Hz), 7.05 (1H, br d, J=9.0 Hz), 7.22-7.41 (2H, m), 7.62 (2H, td, J=8.8, 2.8 Hz), 8.40 (1H, d, J=3.0 Hz), 8.51 (1H, d, J=2.6 Hz).

(238) MS: [M+H-Boc].sup.+252.0.

D) 2-[(1R,2R)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one

(239) To a mixture of tert-butyl [(1R,2R)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]carbamate (4.34 g) and acetic acid (30.9 mL) was added dropwise hydrogen bromide (30% acetic acid solution (ca. 5.1 M), 19.4 mL) at room temperature, and the mixture was stirred at room temperature for 3 hr. The mixture was concentrated under reduced pressure, and the residue was subjected to azeotropy with toluene. To a mixture of the obtained residue, methyl 2-(bromomethyl)-5-(5-(difluoromethyl)-1,3,4-oxadiazol-2-yl)benzoate (4.29 g) and DMF (50 mL) was added dropwise DIPEA (21.5 mL) at 0° C., and the mixture was stirred overnight at room temperature. The mixture was poured into water, and extracted with ethyl acetate. The extract was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (ethyl acetate/hexane), and crystallized from diisopropyl ether/ethyl acetate to give the title compound (3.68 g).

(240) ¹H NMR (300 MHz, DMSO-d₆) δ 4.69 (1H, d, J=18.8 Hz), 4.95 (1H, d, J=18.8 Hz), 5.49 (1H, dd, J=9.0, 5.3 Hz), 5.75 (1H, d, J=9.0 Hz), 6.05 (1H, d, J=5.3 Hz), 7.33-7.79 (5H, m), 7.84 (1H, d, J=8.3 Hz), 8.06 (1H, d, J=1.1 Hz), 8.19-8.26 (1H, m), 8.26-8.31 (1H, m), 8.62 (1H, d, J=3.0 Hz).

(241) MS: [M+H].sup.+ 486.1.

E) 2-[(1R,2R)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 5) and

2-[(1S,2S)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 6)

(242) 2-[(1R,2R)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (3.52 g) was optically resolved by preparative HPLC (column: CHIRALCEL OD, 50 mmID×500 mmL, 20 µm, mobile phase: hexane/ethanol=700/300 (v/v)).

(243) The fraction of the optical isomer having a longer retention time was concentrated under reduced pressure, the residue was dissolved in ethyl acetate, and the solution was passed through short silica gel column. The filtrate was concentrated under reduced pressure, and the residue was crystallized from diisopropyl ether/ethyl acetate. The obtained solid was recrystallized from heptane/ethyl acetate. The obtained solid was further recrystallized from heptane/isopropyl acetate, and dried under reduced pressure at 55-60° C. to give a white solid (1.42 g). 593 mg of the white solid was recrystallized from pentane/isopropyl acetate to give 2-[(1R,2R)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (506 mg, 99% e.e., retention time: 7.77 min (analysis condition, column: CHIRALCEL OD-H, 4.6 mmID×250 mmL, 5 µm, mobile phase: hexane/ethanol/diethylamine=700/300/1 (v/v/v)). The absolute stereochemistry of the compound was determined by X-ray crystallography.

(244) .sup.1H NMR (300 MHz, DMSO-d.sub.6) δ4.68 (1H, d, J=18.4 Hz), 4.95 (1H, d, J=18.8 Hz), 5.49 (1H, dd, J=9.0, 5.6 Hz), 5.75 (1H, d, J=9.0 Hz), 6.04 (1H, d, J=6.0 Hz), 7.34-7.80 (5H, m), 7.84 (1H, d, J=7.9 Hz), 8.05 (1H, d, J=1.1 Hz), 8.23 (1H, dd, J=7.9, 1.5 Hz), 8.27 (1H, t, J=1.5 Hz), 8.61 (1H, d, J=3.0 Hz).

(245) MS: [M+H].sup.+ 486.1

(246) Anal. Calcd for C.sub.23H.sub.15F.sub.4N.sub.5O.sub.3.Math.1H.sub.2O:C, 56.70; H, 3.14; N, 14.37. Found: C, 56.57; H, 3.21; N, 14.32.

(247) powder X-RAY crystal diffraction pattern (diffraction angle 2θ): 6.1°, 9.0°, 15.7°, 16.4°, 18.1°, 19.9°, 21.7°, 22.5°, 24.8°, 25.3°

(248) The fraction of the optical isomer having a shorter retention time was concentrated under reduced pressure, the residue was crystallized from diisopropyl ether/ethyl acetate, and the obtained solid was further recrystallized from heptane/isopropyl acetate, and dried under reduced pressure to give a white solid (1.54 g). 100 mg of the white solid was recrystallized from pentane/isopropyl acetate to give 2-[(1S,2S)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (83 mg, 99% e.e., retention time: 6.43 min (analysis condition, column: CHIRALCEL OD-H, 4.6 mmID×250 mmL, 5 µm, mobile phase: hexane/ethanol/diethylamine=700/300/1 (v/v/v)).

(249) MS: [M+H].sup.+ 486.1

Example 7 and Example 8

6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 7)

6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 8)

A) tert-butyl [2-(6-fluoropyridin-2-yl)-2-oxo-1-(pyridin-2-yl)ethyl]carbamate

(250) To a mixture of tert-butyl [(4-methylbenzene-1-sulfonyl) (pyridin-2-yl)methyl]carbamate (11.9 g), 3-ethyl-5-(2-hydroxyethyl)-4-methylthiazolium bromide (2.48 g) and THF (230 mL) was added 6-fluoropyridine-2-carbaldehyde (4.60 g) under argon atmosphere at room temperature. The mixture was degassed, and replaced with argon gas. To the mixture was added dropwise triethylamine (68.5 mL) at 60° C., and the mixture was stirred overnight under argon atmosphere at 60° C. To the mixture was added water at room temperature, and the mixture was extracted with ethyl acetate. The extract was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (ethyl acetate/hexane) to give the title compound (11.4 g) as a crude product. The obtained compound was used in the next step without further purification.

(251) MS: [M+H].sup.+ 332.1.

B) tert-butyl [(1RS,2RS)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]carbamate (252) To a mixture of tert-butyl [2-(6-fluoropyridin-2-yl)-2-oxo-1-(pyridin-2-yl)ethyl]carbamate (11.4 g), methanol (100 mL) and THF (100 mL) was added sodium borohydride (1.49 g) at 0° C. The mixture was stirred at room temperature for 2.5 hr. To the mixture was added water at 0° C., and the mixture was extracted with ethyl acetate. The separated organic layer was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was crystallized from toluene, collected by filtration, and washed with diisopropyl ether. The obtained solid was recrystallized from hexane/ethyl acetate to give the title compound (6.92 g).

(253) .sup.1H NMR (300 MHz, DMSO-d.sub.6) δ 1.16-1.32 (9H, m), 4.78-4.88 (1H, m), 4.90-5.01 (1H, m), 5.74 (1H, d, J=5.3 Hz), 6.95-7.10 (2H, m), 7.16-7.30 (3H, m), 7.68 (1H, t, J=6.8 Hz), 7.82-7.96 (1H, m), 8.42 (1H, d, J=4.5 Hz).

(254) MS: [M+H].sup.+ 334.1.

C) 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one

(255) To a mixture of tert-butyl [(1RS,2RS)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]carbamate (4.55 g) and acetic acid (80 mL) was added hydrogen bromide (30% acetic acid solution (ca. 5.1 M), 21.4 mL) at room temperature. The mixture was stirred at room temperature for 30 min, concentrated under reduced pressure, and subjected to azeotropy with toluene. To a mixture of the obtained residue and DMF (50 mL) was added dropwise DIPEA (13.6 mL) at 0° C., and then a solution prepared by dissolving methyl 2-(bromomethyl)-5-(5-(difluoromethyl)-1,3,4-oxadiazol-2-yl)benzoate (4.51 g) in DMF (10 mL) was added to the mixture at 0° C. The mixture was stirred at 0° C. to room temperature for 3 days. The mixture was concentrated, and to the residue was added water at room temperature. The mixture was extracted with ethyl acetate. The separated organic layer was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (hexane/ethyl acetate), and crystallized from hexane/ethyl acetate to give the title compound (4.29 g).

(256) .sup.1H NMR (300 MHz, DMSO-d.sub.6) δ 4.68 (1H, d, J=18.8 Hz), 4.98 (1H, d, J=18.8 Hz), 5.39-5.48 (1H, m), 5.71 (1H, d, J=9.0 Hz), 6.11 (1H, d, J=4.0 Hz), 6.95 (1H, dd, J=8.1, 2.4 Hz), 7.32-7.73 (4H, m), 7.77-7.88 (2H, m), 7.89-8.01 (1H, m), 8.06 (1H, d, J=0.9 Hz), 8.23 (1H, dd, J=8.0, 1.6 Hz), 8.58-8.67 (1H, m).

(257) MS: [M+H].sup.+ 468.1.

D) 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 7) and 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 8)

(258) 6-[5-(Difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (4.27 g) was optically resolved by preparative HPLC (column: CHIRALPAK IA, 50 mmID×500 mmL, 20 μ m, mobile phase: hexane/ethanol=200/800 (v/v)).

(259) The fraction of the optical isomer having a shorter retention time was concentrated under reduced pressure, and the residue was recrystallized from heptane/isopropyl acetate to give 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (1.72 g, >99% e.e., retention time: 15.49 min (analysis condition, column: CHIRALPAK IA, 4.6 mmID×250 mmL, 5 μ m, mobile phase: hexane/ethanol=200/800 (v/v)). The absolute stereochemistry of the compound was determined by X-ray crystallography.

(260) .sup.1H NMR (300 MHz, DMSO-d.sub.6) δ 4.68 (1H, d, J=18.8 Hz), 4.98 (1H, d, J=18.8 Hz), 5.43 (1H, dd, J=8.8, 5.8 Hz), 5.71 (1H, d, J=8.7 Hz), 6.10 (1H, d, J=6.0 Hz), 6.95 (1H, dd,

J=7.9, 2.6 Hz), 7.32-7.73 (4H, m), 7.76-7.89 (2H, m), 7.89-8.00 (1H, m), 8.07 (1H, d, J=1.1 Hz), 8.23 (1H, dd, J=7.9, 1.5 Hz), 8.56-8.66 (1H, m).

(261) MS: [M+H].sup.+ 468.1.

(262) Anal. Calcd for C.sub.23H.sub.16F.sub.3N.sub.5O.sub.3Math.2H.sub.2O: C, 58.65; H, 3.51; N, 14.87. Found: C, 58.78; H, 3.75; N, 15.00.

(263) powder X-RAY crystal diffraction pattern (diffraction angle 2θ): 5.7°, 12.1°, 13.0°, 16.7°, 19.0°, 20.2°, 21.8°, 24.2°

(264) The fraction of the optical isomer having a longer retention time was concentrated under reduced pressure, and the residue was recrystallized from heptane/isopropyl acetate to give 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (1.84 g, 99% e.e., retention time: 19.48 min (analysis condition, column: CHIRALPAK IA, 4.6 mmID×250 mmL, 5 μm, mobile phase:

hexane/ethanol=200/800 (v/v)).

(265) MS: [M+H].sup.+ 468.1.

Example 9

2-[(1R,2S)-1-cyclopropyl-2-(4-fluorophenyl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one

A) tert-butyl {(1R)-1-cyclopropyl-2-[methoxy(methyl)amino]-2-oxoethyl}carbamate

(266) To a mixture of (2R)-((tert-butoxycarbonyl)amino) (cyclopropyl)acetic acid (10.2 g) and THF (150 mL) was added CDI (9.22 g) at 0° C., and the mixture was stirred at 0° C. for 1 hr. N-Methoxymethanamine hydrochloride (5.55 g) and DIPEA (12.4 mL) were dissolved in DMF (30 mL), and the obtained solution was added to the mixture at 0° C. The mixture was stirred overnight at room temperature. The mixture was concentrated under reduced pressure. To the residue was added water, and the mixture was extracted with ethyl acetate/THF. The separated organic layer was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (hexane/ethyl acetate) to give the title compound (12.1 g).

(267) .sup.1H NMR (300 MHz, DMSO-d.sub.6) δ 0.21-0.32 (1H, m), 0.33-0.49 (3H, m), 0.93-1.09 (1H, m), 1.36 (9H, s), 3.11 (3H, s), 3.70 (3H, s), 3.90-4.09 (1H, m), 7.06 (1H, d, J=8.3 Hz).

(268) MS: [M+H-Boc].sup.+159.1.

B) tert-butyl [(1R)-1-cyclopropyl-2-(4-fluorophenyl)-2-oxoethyl]carbamate

(269) To a mixture of tert-butyl {(1R)-1-cyclopropyl-2-[methoxy(methyl)amino]-2-oxoethyl}carbamate (8.55 g) and THF (130 mL) was added 4-fluorophenylmagnesium bromide (2 M diethyl ether solution, 42 mL) at 40° C. The mixture was stirred under nitrogen atmosphere at 40° C. for 1 hr. To the mixture was added 4-fluorophenylmagnesium bromide (2 M diethyl ether solution, 3 mL) at 40° C. The mixture was stirred under nitrogen atmosphere at 40° C. for 30 min. The mixture was poured into saturated aqueous ammonium chloride solution at 0° C. To the mixture was added water, and the mixture was extracted with ethyl acetate. The separated organic layer was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (hexane/ethyl acetate) to give the title compound (7.80 g).

(270) .sup.1H NMR (300 MHz, DMSO-d.sub.6) δ 0.19-0.56 (4H, m), 0.91-1.10 (1H, m), 1.13-1.50 (9H, m), 4.09-4.50 (1H, m), 7.35 (2H, t, J=8.8 Hz), 7.46 (1H, br d, J=7.2 Hz), 8.04 (2H, dd, J=8.7, 5.6 Hz).

(271) MS: [M+H-Boc].sup.+194.1.

C) tert-butyl [(1R,2S)-1-cyclopropyl-2-(4-fluorophenyl)-2-hydroxyethyl]carbamate

(272) To a mixture of tert-butyl [(1R)-1-cyclopropyl-2-(4-fluorophenyl)-2-oxoethyl]carbamate (7.80 g), toluene (22 mL) and 2-propanol (23 mL) was added aluminium triisopropoxide (1.09 g) at room temperature. The mixture was stirred at 70° C. 3 hr. The reaction mixture was added to a mixture of 0.5 M hydrochloric acid and ethyl acetate at 0° C., and the mixture was extracted with

ethyl acetate. The separated organic layer was washed with saturated aqueous sodium hydrogencarbonate solution and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure to give the title compound (7.79 g).

(273) ¹H NMR (300 MHz, DMSO-*d*₆) δ -0.24--0.08 (1H, m), 0.09-0.37 (3H, m), 0.95-1.13 (1H, m), 1.26 (9H, s), 3.08-3.25 (1H, m), 4.50 (1H, t, J=5.6 Hz), 5.37 (1H, d, J=4.5 Hz), 6.45 (1H, br d, J=9.4 Hz), 7.09 (2H, t, J=9.0 Hz), 7.29-7.41 (2H, m).

D) (1S,2R)-2-amino-2-cyclopropyl-1-(4-fluorophenyl)ethan-1-ol hydrobromide

(274) To a mixture of tert-butyl [(1R,2S)-1-cyclopropyl-2-(4-fluorophenyl)-2-hydroxyethyl]carbamate (4.79 g) and acetic acid (50 mL) was added hydrogen bromide (25% acetic acid solution, 12.5 mL) at room temperature. The mixture was stirred at room temperature for 30 min. The mixture was concentrated under reduced pressure. The residue was subjected three times to azeotropy with toluene. The residue was suspended in ethyl acetate (10 mL) and diisopropyl ether (40 mL), and the obtained suspension was stirred at room temperature for 30 min. The precipitated solid was collected by filtration, washed with diisopropyl ether/ethyl acetate (4:1), and dried to give the title compound (4.29 g).

(275) ¹H NMR (300 MHz, DMSO-*d*₆) δ -0.41--0.24 (1H, m), 0.12-0.34 (2H, m), 0.34-0.50 (1H, m), 0.77-0.97 (1H, m), 2.59 (1H, dd, J=10.5, 3.0 Hz), 4.94 (1H, br s), 6.15 (1H, d, J=3.8 Hz), 7.09-7.29 (2H, m), 7.37-7.52 (2H, m), 7.89 (3H, br s).

(276) MS: [M+H]⁺ 196.1.

E) 2-[(1R,2S)-1-cyclopropyl-2-(4-fluorophenyl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one

(277) To a mixture of (1S,2R)-2-amino-2-cyclopropyl-1-(4-fluorophenyl)ethan-1-ol hydrobromide (3.65 g), DIPEA (10.5 mL) and DMF (30 mL) was added dropwise a solution prepared by dissolving methyl 2-(bromomethyl)-5-(5-(difluoromethyl)-1,3,4-oxadiazol-2-yl)benzoate (4.17 g) in DMF (12 mL), at room temperature. The mixture was stirred overnight at room temperature. To the mixture was added water at room temperature, and the mixture was extracted with ethyl acetate. The separated organic layer was washed with saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (hexane/ethyl acetate), the obtained solid was dissolved in ethyl acetate, and the solution was filtered to remove the insoluble substance. The filtrate was concentrated under reduced pressure, and the residue was recrystallized from heptane/ethyl acetate to give the title compound (3.75 g).

(278) ¹H NMR (300 MHz, DMSO-*d*₆) δ -0.24--0.11 (1H, m), -0.11--0.01 (1H, m), 0.28-0.51 (2H, m), 1.31-1.48 (1H, m), 3.51 (1H, dd, J=10.2, 4.1 Hz), 4.78 (1H, d, J=19.2 Hz), 4.88 (1H, d, J=19.2 Hz), 5.00 (1H, t, J=4.5 Hz), 5.74 (1H, d, J=4.9 Hz), 7.05-7.21 (2H, m), 7.35-7.78 (3H, m), 7.90 (1H, d, J=8.3 Hz), 8.21 (1H, d, J=0.8 Hz), 8.29 (1H, dd, J=7.9, 1.9 Hz).

(279) MS: [M+H]⁺ 430.1.

(280) Anal. Calcd for C₂₂H₁₈F₃N₃O₃: C, 61.28; H, 4.25; N, 9.75. Found: C, 61.28; H, 4.48; N, 9.77.

(281) powder X-RAY crystal diffraction pattern (diffraction angle 2θ): 6.9°, 13.9°, 16.9°, 18.6°, 20.2°, 20.9°, 22.2°, 22.7°, 25.6°, 28.1°

(282) The compounds of Examples are shown in the following tables. MS in the tables means actual measured value. The compounds of Examples 10-150 in the following tables were produced according to the methods described in the above-mentioned Examples, or methods analogous thereto.

(283) TABLE-US-00001 TABLE 1-1 EXAMPLE IUPAC NAME Structure MS
1 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one
2 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one
3 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(5-

fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 468.1 4 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 468.2 5 2-[(1R,2R)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one  embedded image 486.1 6 2-[(1S,2S)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one  embedded image 486.1 7 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 468.2

(284) TABLE-US-00002 TABLE 1-2 EXAMPLE IUPAC NAME Structure MS 8 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 468.2 9 2-[(1R,2S)-1-cyclopropyl-2-(4-fluorophenyl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one 0  embedded image 430.3 10 2-[(1R,2S)-1-cyclopropyl-2-(3-fluorophenyl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one  embedded image 430.1 11 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2S)-2-hydroxy-1,2-diphenylethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 446.2 12 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-hydroxy-1,2-diphenylethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 448.2 13 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-hydroxy-1,2-diphenylethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 448.2 14 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2R)-2-hydroxy-1,2-diphenylethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 448.2

(285) TABLE-US-00003 TABLE 1-3 EXAMPLE IUPAC NAME Structure MS 15 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-hydroxy-2-(pyridin-2-yl)-1-(3,4,5-trifluorophenyl)ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 503.1 16 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(3-fluorophenyl)-1-(4-fluorophenyl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 484.2 17 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-hydroxy-2-phenyl-1-[6-(trifluoromethyl)pyridin-3-yl]ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 517.2 18 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(4-fluorophenyl)-2-hydroxy-1-[6-(trifluoromethyl)pyridin-3-yl]ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 535.1 19 2-[(1RS,2SR)-2-cyclopropyl-1-(4-fluorophenyl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one 0  embedded image 428.2 20 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-1-(3-fluorophenyl)-2-hydroxy-2-phenylethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 466.1 21 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-1-(3-fluorophenyl)-2-(4-fluorophenyl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 484.1

(286) TABLE-US-00004 TABLE 1-4 EXAMPLE IUPAC NAME Structure MS 22 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(4-fluorophenyl)-2-hydroxy-1-phenylethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 466.1 23 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(3-fluorophenyl)-2-hydroxy-1-phenylethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 466.1 24 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-hydroxy-2-(pyridin-2-yl)-1-[3-(trifluoromethyl)phenyl]ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 517.0 25 2-[(1RS,2SR)-2-(4,4-difluorocyclohexyl)-2-hydroxy-1-phenylethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one  embedded image 490.1 26 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-hydroxy-2-(pyridin-2-yl)-1-[4-(trifluoromethyl)phenyl]ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 517.1 27 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-1-(2-fluorophenyl)-2-hydroxy-2-phenylethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 466.1 28 6-[5-

(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-1-(2-fluorophenyl)-2-(4-fluorophenyl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 484.1

(287) TABLE-US-00005 TABLE 1-5 EXAMPLE IUPAC NAME Structure MS 29 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(2-fluorophenyl)-1-(4-fluorophenyl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one 0  embedded image 484.1 30 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-1-(4-fluorophenyl)-2-hydroxy-2-(oxan-4-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 474.2 31 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-1-(2,4-difluorophenyl)-2-hydroxy-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 485.1 32 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-1-(2,4-difluorophenyl)-2-(4-fluorophenyl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 500.1 33 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-1-(2,4-difluorophenyl)-2-(4-fluorophenyl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 500.1 34 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(4-fluorophenyl)-1-(6-fluoropyridin-3-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 485.1 35 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(4-fluorophenyl)-1-(6-fluoropyridin-3-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 485.1

(288) TABLE-US-00006 TABLE 1-6 EXAMPLE IUPAC NAME Structure MS 36 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(4-fluorophenyl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 467.1 37 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(4-fluorophenyl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 467.1 38 2-[(1R,2R)-1-cyclopropyl-2-hydroxy-2-(pyridin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one  embedded image 413.2 39 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-hydroxy-2-(pyridin-2-yl)-1-[2-(trifluoromethyl)phenyl]ethyl]-2,3-dihydro-1H-isoindol-1-one 0  embedded image 517.1 40 3-[(1RS,2SR)-1-{6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-1-oxo-1,3-dihydro-2H-isoindol-2-yl}-2-hydroxy-2-phenylethyl]benzonitrile  embedded image 471.2 41 4-[(1RS,2SR)-1-{6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-1-oxo-1,3-dihydro-2H-isoindol-2-yl}-2-hydroxy-2-phenylethyl]benzonitrile  embedded image 471.1 42 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-hydroxy-1-(1-methyl-1H-pyrazol-4-yl)-2-phenylethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 452.1

(289) TABLE-US-00007 TABLE 1-7 EXAMPLE IUPAC NAME Structure MS 43 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(4-fluorophenyl)-2-hydroxy-1-(1-methyl-1H-pyrazol-4-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 468.2 44 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(6-ethoxypyridin-3-yl)-2-hydroxy-1-phenylethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 493.1 45 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-1-(4-fluorophenyl)-2-hydroxy-2-phenylethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 464.2 46 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-1-(4-fluorophenyl)-2-hydroxy-2-phenylethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 464.2 47 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-2-hydroxy-1-phenyl-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 449.2 48 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-2-hydroxy-1-phenyl-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 449.2 49 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-1-(4-fluorophenyl)-2-hydroxy-2-(pyridin-3-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2) 0  embedded image 467.1

(290) TABLE-US-00008 TABLE 1-8 EXAMPLE IUPAC NAME Structure MS 50 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-1-(3,4-difluorophenyl)-2-hydroxy-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 485.1 51 6-

[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-1-(3,4-difluorophenyl)-2-hydroxy-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 485.1 52 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-1-(4-fluorophenyl)-2-hydroxy-2-(pyridin-3-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 467.1 53 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(3,4-difluorophenyl)-1-(6-ethoxypyridin-3-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 529.1 54 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-1-(4-fluorophenyl)-2-hydroxy-2-[6-(trifluoromethyl)pyridin-3-yl]ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 533.1 55 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S)-1-(4-fluorophenyl)-2-hydroxy-2-[6-(trifluoromethyl)pyridin-3-yl]ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 533.2 56 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(3,4-difluorophenyl)-1-(6-fluoropyridin-3-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 501.2

(291) TABLE-US-00009 TABLE 1-9 EXAMPLE IUPAC NAME Structure MS 57 2-[(1R,2R)-1-cyclopropyl-2-(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one  embedded image 431.2 58 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-1-(4-fluorophenyl)-2-(6-fluoropyridin-3-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 483.2 59 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-1-(5-fluoropyridin-2-yl)-2-hydroxy-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1) 0  embedded image 468.2 60 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-2-(5-fluoropyridin-2-yl)-2-hydroxy-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 468.2 61 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3-fluorophenyl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 467.1 62 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3-fluorophenyl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 467.1 63 2-[(1R,2S)-1-cyclopropyl-2-(6-fluoropyridin-3-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one  embedded image 429.2

(292) TABLE-US-00010 TABLE 1-10 EXAMPLE IUPAC NAME Structure MS 64 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2S)-2-(3-fluorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 472.2 65 2-[(1RS,2RS)-2-(5-chloropyridin-2-yl)-1-cyclopropyl-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one  embedded image 447.1 66 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-1-(5-fluoropyridin-2-yl)-2-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 486.1 67 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-1-(5-fluoropyridin-2-yl)-2-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 486.1 68 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3-fluorophenyl)-2-hydroxy-1-(pyrimidin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 468.1 69 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3-fluorophenyl)-2-hydroxy-1-(pyrimidin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2) 0  embedded image 468.2 70 2-[(1R*,2S*)-2-(3-chlorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 488.1

(293) TABLE-US-00011 TABLE 1-11 EXAMPLE IUPAC NAME Structure MS 71 2-[(1R*,2S*)-2-(3-chlorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 488.1 72 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2S)-2-(4-fluorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 472.2 73 2-[(1R*,2R*)-1-cyclopropyl-2-hydroxy-2-(pyrimidin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 414.2 74 2-[(1R*,2R*)-1-

cyclopropyl-2-hydroxy- 2-(pyrimidin-2-yl)ethyl]-6-[5- (difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 414.2 75 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-1-(6-fluoropyridin-2-yl)-2-hydroxy-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 468.2 76 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-1-(6-fluoropyridin-2-yl)-2-hydroxy-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 468.2 77 2-[(1R*,2S*)-1-cyclopropyl-2-hydroxy- 2-[6-(trifluoromethyl)pyridin-3-yl] ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 479.1

(294) TABLE-US-00012 TABLE 1-12 EXAMPLE IUPAC NAME Structure MS 78 2-[(1R*,2S*)-1-cyclopropyl-2-hydroxy- 2-[6-(trifluoromethyl)pyridin-3-yl] ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 479.1 79 2-[(1R*,2R*)-1-cyclopropyl-2-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1) 00  embedded image 431.2 80 2-[(1R*,2R*)-1-cyclopropyl-2-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2) 01  embedded image 431.2 81 2-[(1R*,2S*)-1-cyclopropyl-2-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1) 02  embedded image 491.1 82 2-[(1R*,2R*)-1-cyclopropyl-2-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2) 03  embedded image 507.2 83 2-[(1R*,2S*)-1-cyclopropyl-2-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1) 04  embedded image 488.1 84 2-[(1R*,2S*)-1-cyclopropyl-2-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2) 05  embedded image 488.2

(295) TABLE-US-00013 TABLE 1-13 EXAMPLE IUPAC NAME Structure MS 85 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3,4-difluorophenyl)-2-hydroxy-1-(pyrazin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1) 06  embedded image 486.1 86 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3,4-difluorophenyl)-2-hydroxy-1-(pyrazin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2) 07  embedded image 486.1 87 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3,4-difluorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1) 08  embedded image 490.2 88 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3,4-difluorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2) 09  embedded image 490.1 89 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3-fluorophenyl)-1-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1) 0  embedded image 485.1 90 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3-fluorophenyl)-1-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 485.0 91 2-[(1R*,2S*)-2-(3-chlorophenyl)-2-hydroxy-1-(pyrazin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 484.1

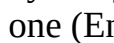
(296) TABLE-US-00014 TABLE 1-14 EXAMPLE IUPAC NAME Structure MS 92 2-[(1R*,2S*)-2-(3-chlorophenyl)-2-hydroxy-1-(pyrazin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 484.1 93 2-[(1R*,2S*)-2-(3-chlorophenyl)-2-hydroxy-1-(pyrazin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 484.1 94 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3,4-difluorophenyl)-2-hydroxy-1-(pyrimidin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 486.1 95 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3,4-difluorophenyl)-2-hydroxy-1-(pyrimidin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 486.1 96

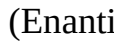
2-[(1R*,2S*)-2-(4-chlorophenyl)-2- hydroxy-1-(pyrazin-2-yl)ethyl]-6-[5- (difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 484.1 97 2-[(1R*,2S*)-2-(4-chlorophenyl)-2- hydroxy-1-(pyrazin-2-yl)ethyl]-6-[5- (difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 484.1 98 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2- yl]-2-[(1R*,2S*)-2-(4-fluorophenyl)-2- hydroxy-1-(pyrazin-2-yl)ethyl]-2,3- dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 468.2

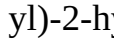
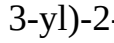
(297) TABLE-US-00015 TABLE 1-15 EXAMPLE IUPAC NAME Structure MS 99 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2- yl]-2-[(1R*,2S*)-2-(4-fluorophenyl)-2- hydroxy-1-(pyrazin-2-yl)ethyl]-2,3- dihydro-1H-isoindol-1-one (Enantiomer 2) 0  embedded image 468.2 100 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2- yl]-2-[(1R*,2S*)-2-(3-fluorophenyl)-2- hydroxy-1-(pyrazin-2-yl)ethyl]-2,3- dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 468.2 101 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2- yl]-2-[(1R*,2S*)-2-(3-fluorophenyl)-2- hydroxy-1-(pyrazin-2-yl)ethyl]-2,3- dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 468.2 102 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2- yl]-2-[(1R*,2S*)-2-hydroxy-2-phenyl- 1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H- isoindol-1-one (Enantiomer 1)  embedded image 449.2 103 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2- yl]-2-[(1R*,2S*)-2-hydroxy-2-phenyl- 1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H- isoindol-1-one (Enantiomer 2)  embedded image 449.2 104 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2- yl]-2-[(1R*,2R*)-2-hydroxy-1-(pyridin- 2-yl)-2-[5-(trifluoromethyl)pyridin-2- yl]ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 518.2 105 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2- yl]-2-[(1R*,2R*)-2-hydroxy-1-(pyridin- 2-yl)-2-[5-(trifluoromethyl)pyridin-2- yl]ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 518.2

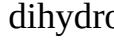
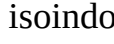
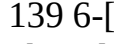
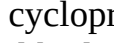
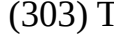
(298) TABLE-US-00016 TABLE 1-16 EXAMPLE IUPAC NAME Structure MS 106 2-[(1RS,2SR)-2-(3-chloro-5- fluorophenyl)-2-hydroxy-1-(oxan-4- yl)ethyl]-6-[5-(difluoromethyl)-1,3,4- oxadiazol-2-yl]-2,3-dihydro-1H- isoindol-1-one  embedded image 506.1 107 2-[(1RS,2SR)-2-(3-chloro-4- fluorophenyl)-2-hydroxy-1-(oxan-4- yl)ethyl]-6-[5-(difluoromethyl)-1,3,4- oxadiazol-2-yl]-2,3-dihydro-1H- isoindol-1-one  embedded image 506.2 108 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2- yl]-2-[(1R*,2S*)-2-(3,4-difluorophenyl)- 2-hydroxy-1-(pyridazin-3-yl)ethyl]-2,3- dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 486.1 109 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2- yl]-2-[(1R*,2S*)-2-(3,4-difluorophenyl)- 2-hydroxy-1-(pyridazin-3-yl)ethyl]-2,3- dihydro-1H-isoindol-1-one (Enantiomer 2) 0  embedded image 486.1 110 2-[(1R*,2S)-2-(4-chlorophenyl)-2- hydroxy-1-(pyridazin-3-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 484.1 111 2-[(1R*,2S*)-2-(4-chlorophenyl)-2- hydroxy-1-(pyridazin-3-yl)ethyl]-6-[5- (difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 484.1 112 2-[(1R*,2R*)-2-(5-chloropyridin-2-yl)- 2-hydroxy-1-(pyridin-2-yl)ethyl]-6-[5- (difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 484.1

(299) TABLE-US-00017 TABLE 1-17 EXAMPLE IUPAC NAME Structure MS 113 2-[(1R*,2R)-2-(5-chloropyridin-2-yl)- 2-hydroxy-1-(pyridin-2-yl)ethyl]-6-[5- (difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 484.1 114 2-[(1R*,2R*)-2-(6-chloropyridin-2-yl)- 2-hydroxy-1-(pyridin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  embedded image 484.1 115 2-[(1R*,2R*)-2-(6-chloropyridin-2-yl)- 2-hydroxy-1-(pyridin-2-yl)ethyl]-6-[5- (difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 484.0 116 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2- yl]-2-[(1RS,2SR)-2-hydroxy-1-(oxan-4- yl)-2-[3-(trifluoromethyl)phenyl]ethyl]- 2,3-dihydro-1H-isoindol-1-one  embedded image 522.1 117 2-[(1R*,2R*)-1,2-bis(6-fluoropyridin-2- yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  embedded image 486.1 118 2-[(1R*,2R*)-1,2-bis(6-fluoropyridin-2- yl)-2-hydroxyethyl]-6-[5-

(difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  486.1 119 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-2-hydroxy-1-(pyridin-2-yl)-2-[6-(trifluoromethyl)pyridin-2-yl]ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  518.1

(300) TABLE-US-00018 TABLE 1-18 EXAMPLE IUPAC NAME Structure MS 120 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-2-hydroxy-1-(pyridin-2-yl)-2-[6-(trifluoromethyl)pyridin-2-yl]ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  518.1 121 2-[(1R*,2S)-2-(3-chlorophenyl)-2-hydroxy-1-(pyridazin-3-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  484.1 122 2-[(1R*,2S*)-2-(3-chlorophenyl)-2-hydroxy-1-(pyridazin-3-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  484.1 123 2-[(1R*,2R*)-2-(4-chloropyridin-2-yl)- 2-hydroxy-1-(pyridin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  484.1 124 2-[(1R*,2R*)-2-(4-chloropyridin-2-yl)- 2-hydroxy-1-(pyridin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  484.1 125 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R)-2-(5-fluoropyridin-2-yl)-1-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  486.1 126 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-2-(5-fluoropyridin-2-yl)-1-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  486.1

(301) TABLE-US-00019 TABLE 1-19 EXAMPLE IUPAC NAME Structure MS 127 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-hydroxy-2-phenyl-1-(pyrazin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  450.1 128 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-hydroxy-2-phenyl-1-(pyrazin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  450.1 129 2-[(1R*,2R*)-2-(6-chloropyridin-2-yl)-1-cyclopropyl-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  447.1 130 2-[(1R*,2R*)-2-(6-chloropyridin-2-yl)-1-cyclopropyl-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  447.1 131 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-1-(4-fluorophenyl)-2-(6-fluoropyridin-3-yl)-2-hydroxyethyl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)  485.1 132 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-1-(4-fluorophenyl)-2-(6-fluoropyridin-3-yl)-2-hydroxyethyl]- 2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)  485.1 133 2-[(1RS,2SR)-2-(3-chlorophenyl)-2-hydroxy-1-(pyrimidin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one  484.1

(302) TABLE-US-00020 TABLE 1-20 EXAMPLE IUPAC NAME Structure MS 134 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(4-fluorophenyl)-2-hydroxy-1-(pyrimidin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one  468.2 135 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one  450.1 136 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-(3-fluorophenyl)-2-hydroxy-1-(pyrimidin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one  468.2 137 2-[(1R,2R)-1-cyclopropyl-2-(4-fluorophenyl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one  430.2 138 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1SR,2SR)-1-(4-fluorophenyl)-2-hydroxy-2-phenylethyl]-2,3-dihydro-1H-isoindol-1-one  446.1 139 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-(2-fluorophenyl)-2-hydroxy-1-phenylethyl]-2,3-dihydro-1H-isoindol-1-one  466.1 140 2-[(1RS,2RS)-2-cyclopropyl-1-(4-fluorophenyl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]- 2,3-dihydro-1H-isoindol-1-one  430.2

(303) TABLE-US-00021 TABLE 1-21 EXAMPLE IUPAC NAME Structure MS 141 6-[5-

(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-(4-fluorophenyl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 467.1 142 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-1-(4-fluorophenyl)-2-hydroxy-2-[6-(trifluoromethyl)pyridin-3-yl]ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 533.1 143 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-hydroxy-2-phenyl-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 449.2 144 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-1-(5-fluoropyridin-2-yl)-2-hydroxy-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 468.2 145 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1SR,2SR)-2-(3-fluorophenyl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 467.1 146 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1SR,2SR)-1-(2,4-difluorophenyl)-2-(4-fluorophenyl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 500.1 147 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-(3-fluorophenyl)-1-(4-fluorophenyl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 484.1

(304) TABLE-US-00022 TABLE 1-22 EXAMPLE IUPAC NAME Structure MS 148 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(4-fluorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one  embedded image 472.2 149 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(3-fluorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one 0  embedded image 472.2 150 2-[(1R,2S)-1-cyclopropyl-2-(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one  embedded image 431.2

Experimental Example 1 HDAC6 Enzyme Inhibitory Assay

(305) HDAC6 enzyme prepared by transducing full length HDAC6 gene into Sf-9 insect cells and purifying by GST affinity column were purchased from SignalChem. Using this enzyme, HDAC6 enzyme inhibitory activity of the compound of the present invention were evaluated. Enzymes were used after preserved at -70° C. HDAC6 enzyme inhibitory activity of the compound of the present invention was measured using HDAC-Glo™ I/II Assay kit (Promega) according to the following experimental method. The test compound diluted with assay buffer (24 mM Tris-HCl (pH 7.5), 1 mM MgCl₂, 0.35 mM KCl, 135 mM NaCl, 0.6 mM Glutathione, 0.01% Tween-20) was added to a 384-well plate by each 2 μL. Then, HDAC6 enzyme solution diluted with assay buffer was added thereto by each 4 μL, and the plate was incubated at room temperature for 60 min. After incubated, HDAC substrate-Developer solution prepared according to Promega protocol attached to the assay kit was added to the 384-well plate by each 2 μL, and the enzyme reaction was started. After reacting at room temperature for 20 min, luminescence level was measured using plate reader Envision (PerkinElmer). The inhibitory activity of each compound of Example was calculated as a relative activity value when luminescence level in wells without enzyme is considered as 100% inhibition. The results are shown in Table 2-1 and Table 2-2.

(306) TABLE-US-00023 TABLE 2-1 Ex. HDAC6 inhibitory No. rate (%) (10 μM)

1	101	2	100	3
101	4	100	5	99
6	100	7	102	8
99	9	100	10	100
11	99	12	100	13
99	14	100	15	99
16	99	17	100	18
99	19	99	20	100
21	100	22	99	23
100	24	99	25	100
26	99	27	101	28
99	29	99	30	99
31	97	32	99	33
100	34	100	35	100
36	101	37	100	38
100	39	101	40	100
41	100	42	100	43
100	44	100	45	100
46	100	47	99	48
100	49	99	50	99
51	99	52	100	53
100	54	100	55	98
56	100	57	100	58
100	59	100	60	99
61	100	62	100	63
100	64	100	65	98
66	100	67	99	68
101	69	100	70	101
71	101	72	100	73
100	74	96	75	99
76	100	77	100	78
99	79	99	80	100
81	100	82	100	83
101	84	100	85	100
86	100	87	100	88
99	89	100	90	100

(307) TABLE-US-00024 TABLE 2-2 Ex. HDAC6 inhibitory No. rate (%) (10 μM)

91	100	92	100
93	100	94	100
95	99	96	100
97	99	98	100
99	100	100	100
101	100	102	100
103	100	104	100
105	99	106	100
107	101	108	100
109	100	110	100
111	99	112	99
113	100	114	99
115	100	116	101
117	101	118	102
119	101	120	101
121	101	122	100
123	101	124	100
125	101	126	101
127	99	128	99
129	100	130	98
131	99	132	99
133	100	134	101
135	99	136	99
137	100	138	99
139	100	140	98

(308) As is clear from Table 2-1 and Table 2-2, the compound of the present invention has an excellent HDAC6 inhibitory activity.

Experimental Example 2 Increase in Acetylated Tubulin in Brain

(309) A drug was suspended in 0.5% methyl cellulose to prepare a suspension for administration, having a concentration shown in Table 3. The suspension was orally administered into 8 to 10-week old male C57BL/6J mouse, and after the passage of the time shown in Table 3, the brain was taken out, and hippocampus was obtained. The hippocampus was homogenized under RIPA extract (FUJIFILM Wako Pure Chemical Corporation) to which protease inhibitor (Thermo Fisher Scientific) and phosphatase inhibitor (Thermo Fisher Scientific) were added, and centrifuged in 15,000 g for 15 min to prepare a protein extract. The acetylated tubulin and total tubulin in the extract were detected by the following ELISA method. Tubulin antibody (Sigma) was immobilized on 96-well plate (Microlite2+, Thermo Fisher Scientific), and the plate was stored at 4° C. Next day, the plate was washed four times with PBS-T, and blocking buffer (1% BSA/PBS-T) was added thereto, and the reaction was carried out at 37° C. for 2 hr. The plate was washed four times with PBS-T, and the above-mentioned protein extract was added thereto to capture the tubulin on the antibody. The plate was stored at 37° C. for 2 hr, and washed four times with PBS-T. Anti-acetylated tubulin antibody or total tubulin antibody (Cell Signaling Technology), each diluted with blocking buffer, was added thereto, and the reaction was carried out at 37° C. for 1 hr. Then, the plate was washed four times with PBS-T, and the reaction was carried out using anti-mouse HRP (Cell Signaling Technology) at 37° C. for 30 min. The plate was washed, HRP substrate was added thereto, and the luminescence was measured using a plate reader. The acetylated tubulin and total tubulin were quantified from logistics curve based on dilution series using the mouse hippocampus tissue extract, and the amount of the acetylated tubulin relative to the total tubulin was calculated, and evaluated as an relative tubulin acetylation level. The test was performed using SAS system 8. After F test for vehicle group and drug group, the significant difference between the two groups was analyzed by Student t-test or Welch test (indicated as *p<0.05, **p<0.01, n.s.: not significant). The graph was indicated by the mean±standard error. The results are shown in FIGS. 1 to 3. In the mouse into which the compound of Example 1, 3 or 5 was administered, increase in significant tubulin acetylation level was observed.

(310) TABLE-US-00025 TABLE 3 time (h) after Ex. concentration dose administration of C57BL/6J C573L/6J No. (mg/mL) (mg/kg) test compound old number 1 0.01, 0.1, 0.1, 1, 4 9 6 1, 10 10, 100 3 0.01, 0.1, 0.1, 1, 4 8 6 1, 10 10, 100 5 0.03, 0.1 0.3, 1 4 10 6

Experimental Example 3 Cognitive Improvement Action in Neurodegenerative Disease Model Mouse

(311) A drug was suspended in 0.5% methyl cellulose to prepare a suspension for administration. The suspension was repeatedly administered into six month-old male P301S mutant human tau (4R1N) Tg mouse once a day for three months, and novel object recognition test was performed by the following method. The test is comprised of memory acquisition trial of object on the first day, and memory retention trial of familiar object and novel object on the second day. Drug was administered into the mouse 1 hr before the trials on both days. For acquisition trial, The mouse was put in a test box (30 cm×30 cm×25 cm) in which two identical objects were placed, and contact frequency and contact duration with the objects for 5 min were measured under 50 lux. Herein, the contact means a behavior getting a sniff of an object (Sniffing). Next day, one of the objects was replaced with new object, and contact frequency and contact duration with each object for 5 min were measured. Novelty discrimination index (NDI) as an index of cognitive function was calculated by novel object contact frequency/(novel object contact duration+familiar object contact duration) %. The test was performed using SAS system 8. After F test for vehicle administration wild-type group and vehicle administration Tg group, the significant difference between the two groups was analyzed by Student t-test or Welch test (indicated as **p<0.01). In

addition, Bartlett test for homogeneity of variance was performed for vehicle administration or drug administration Tg group, and the significant difference was analyzed by one-tailed Williams test (indicated as $p < 0.005$). Each group consisted of 14 or 15 mice. The graph was indicated by the mean $\pm$ standard error. The results are shown in FIGS. 4 and 5.

Experimental Example 4 Toxicity Test in Rat

(312) A Two-week toxicity test in rats is conducted using the following test system.

(313) Test System:

(314) Rat, Sprague-Dawley [CrI:CD(SD)]

(315) Administration Method/Period:

(316) Oral administration, which is the scheduled route for clinical application, is selected as the administration route, and the administration period is 2 weeks. The administration frequency is once a day (7 days/week), which is generally used in repeated dose studies. For oral administration to rodents, the administration method is general oral gavage. The dose is 5 mL/kg, and it is administered by oral gavage using a flexible gastric tube (between 08:00 and 15:00). The vehicle is similarly administered to the vehicle control group. The dose of each animal is calculated based on the most recent body weight.

(317) Vehicle and Preparation Method:

(318) 0.5 w/v % methylcellulose solution and the required amount of methylcellulose (METOLOSE SM-100) are weighed and gradually added to an appropriate amount of warm water for injection with stirring to disperse the methylcellulose. This is cooled to dissolve the methylcellulose, and water for injection is further added thereto to give a 0.5% solution.

(319) A hematology protocol is described below as an example of toxicity assessment.

(320) Materials:

(321) At the time of planned autopsy, all animals in the test group that had been fasted overnight from the previous day (about 16 to 21 hours) are subjected to laparotomy under isoflurane inhalation anesthesia, and the blood (about 1 mL) is collected from the abdominal aorta into a blood sampling bottle containing EDTA-2K. The obtained blood is tested for the items and by the methods listed below.

(322) Used Equipment:

(323) Automatic blood cell counter (ADVIA2120i, Siemens Healthcare Diagnostics)

(324) Test Items (Measurement Method):

(325) Red blood cell count (2-angle laser flow cytometry method) Hematocrit value (calculated from red blood cell count and MCV) Hemoglobin concentration (modified cyanmethemoglobin method) Average red blood cell hemoglobin amount (calculated from MCH, red blood cell count and hemoglobin concentration) Average red blood cell hemoglobin concentration (calculated from MCHC, red blood cell count, MCV and hemoglobin concentration) Average red blood cell volume (MCV, 2-angle laser flow cytometry method) Red blood cell volume distribution width (RDW, 2-angle laser flow cytometry method) Reticulocyte count (absolute count, laser flow cytometry by RNA staining) Platelet count (2-angle laser flow cytometry method) White blood cell count (2-angle laser flow cytometry method) White blood cell classification (absolute count, flow cytometry by peroxidase staining+2-angle laser flow cytometry) Erythrocyte morphology flag (MICRO: microcytic red blood cells, MACRO: macrocytic red blood cells, HYPO: hypochromic red blood cells, HYPER: hyperchromic red blood cells, ANISO: red blood cell size difference, HCVAR: hemoglobin concentration difference, RBCF: crushed red blood cells)

Experimental Example 5 Bacterial Reversion Assay

Materials and Methods

(326) Test Strains:

(327) *Salmonella typhimurium* TA100 or TA98

(328) Metabolic Activation System:

(329) S9 mix for Ames test prepared from livers of 7-week-old male Sprague-Dawley rats treated

with phenobarbital and 5,6-benzoflavone

(330) Doses:

(331) 78.1, 156.3, 312.5, 625, 1250, 2500 or 5000 µg/plate

(332) Test Method:

(333) After the bacterial cells are cultured at 37° C. for 8 hours, the optical density (at 660 nm) of cell suspension is measured by spectrophotometer to determine if bacteria grows well. The cell numbers calculated from the optical density are confirmed to be within the acceptable range (cell number: $\geq 1 \times 10^9$ /mL). Cells are exposed to test article using the pre-incubation method (37° C., 20 minutes) in the presence or absence of S9 mix. After mixing with top agar, cells are overlaid on plates. The plates are incubated at 37° C. for 48 hours. The top agar, solvent, and the test article solution are inspected for bacterial contamination.

(334) Criteria for Mutagenicity:

(335) The test article is determined to be positive if the test article induced a dose-dependent increase in the number of the revertant colonies to a level equal to or greater than 2 times of the mean negative control value in any test strain either in the presence or absence of S9 mix.

Experimental Example 6 In Vitro Micronucleus Assay

Materials and Methods

(336) Test System:

(337) Human lymphoblastoid cells derived TK6 cells

(338) Culture Condition:

(339) The cells are cultured in a 6-well plastic plate at 37° C. and 5% CO₂ with RPMI1640 supplemented with 10 vol % heat-inactivated horse serum, 2 mmol/L sodium pyruvate, 100 unit/mL penicillin and 100 µg/mL streptomycin.

(340) Metabolic Activation System:

(341) Rat liver S9 mix prepared from the livers of 7-week-old male Sprague-Dawley rats treated with phenobarbital and 5,6-benzoflavone.

(342) Treatment Conditions:

(343) (1) 3-hour treatment with S9 mix, followed by 21-hour culture, (2) 24-hour continuous treatment without S9 mix

Doses:

(344) 450.0 µg/mL to 12.6 µg/mL

(345) Duplicate cultures are prepared for each concentration.

(346) Cytotoxicity:

(347) Cell numbers are counted, and the relative population doubling (RPD) as a cytotoxic evaluation index is calculated using the following formula:

(348)

$$\text{Populationdoubling(PD)} = \frac{\log_{10}([\text{cellnumberattheendofculture}] / [\text{cellnumberatthestartoftreatment}^*])}{\log_{10} 2} * : 1 \times 10^5 \text{ cells / mL}$$

$$\text{RPD(\%)} = \frac{\text{PDforthetestarticlegroup}}{\text{PDfortheconcurrentnegativecontrolgroup}} \times 100$$

Micronucleus Observation:

(349) For each treatment condition, the highest concentration for micronucleus analysis is selected based on precipitation of test articles and cytotoxicity, and two lower consecutive concentrations are examined. The number of micronucleated cells among 4,000 mononuclear cells with cytoplasm is counted using a microscope.

(350) Statistical Analysis:

(351) Fisher's exact test is performed at upper-tailed significance levels of 5% and 1% in order to compare the incidence of micronucleated cells in the test article groups or the positive control group with that of the negative control group for each treatment condition. If Fisher's exact test shows statistically significance in the test article groups, the exact Cochran-Armitage trend test is performed.

(352) Judgment Criteria:

(353) The test article is judged to be positive if the incidence of micronucleated cells between any test article groups satisfied both criteria (1) and (2) shown below. (1) A significant difference in the incidence of micronucleated cells between the negative control group is detected in the statistical analysis and dose-dependent increase is also detected in the statistical analysis. (2) The incidence of micronucleated cells is more than the historical control range (mean+2SD). Otherwise, the test article is considered to be negative.

Formulation Example 1 (Production of Capsule)

(354) TABLE-US-00026 1) compound of Example 1 30 mg 2) fine powder cellulose 10 mg 3) lactose 19 mg 4) magnesium stearate 1 mg Total 60 mg

(355) 1), 2), 3) and 4) are mixed and filled in a gelatin.

(356) TABLE-US-00027 1) compound of Example 1 30 g 2) lactose 50 g 3) cornstarch 15 g 4) calcium carboxymethylcellulose 44 g 5) magnesium stearate 1 g 1000 tablets total 140 g

(357) The total amount of 1), 2) and 3) and 4) (30 g) is kneaded with water, vacuum dried, and sieved. The sieved powder is mixed with 4) (14 g) and 5) (1 g), and the mixture is punched by a tableting machine, whereby 1000 tablets containing 30 mg of the compound of Example 1 per tablet are obtained.

INDUSTRIAL APPLICABILITY

(358) The compound of the present invention has a HDAC inhibitory activity, and may be useful for the treatment of central nervous system diseases including neurodegenerative diseases (Alzheimer's disease, progressive supranuclear palsy, etc.) and the like.

(359) This application is based on patent applications No. 2019-177815 filed on Sep. 27, 2019 in Japan, the contents of which are encompassed in full herein.

Claims

1. A compound represented by the formula (I): ##STR00172## wherein R^{sup.1} is (1) a C_{sub}.3-10 cycloalkyl group optionally substituted by 1 to 3 halogen atoms, (2) a C_{sub}.6-14 aryl group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a cyano group, and (iii) a C_{sub}.1-6 alkyl group optionally substituted by 1 to 3 halogen atoms, (3) a 5- or 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a C_{sub}.1-6 alkyl group optionally substituted by 1 to 3 halogen atoms, and (iii) a C_{sub}.1-6 alkoxy group, or (4) a 3- to 8-membered monocyclic non-aromatic heterocyclic group, and R^{sup.2} is (1) a C_{sub}.3-10 cycloalkyl group optionally substituted by 1 to 3 halogen atoms, (2) a C_{sub}.6-14 aryl group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, and (ii) a C_{sub}.1-6 alkyl group optionally substituted by 1 to 3 halogen atoms, (3) a 5- or 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a C_{sub}.1-6 alkyl group optionally substituted by 1 to 3 halogen atoms, and (iii) a C_{sub}.1-6 alkoxy group, or (4) a 3- to 8-membered monocyclic non-aromatic heterocyclic group, or a salt thereof.

2. The compound or salt according to claim 1, which is represented by the formula (I'):
##STR00173## wherein each symbol is as defined in claim 1.

3. The compound or salt according to claim 1, wherein R^{sup.1} is (1) a C_{sub}.3-10 cycloalkyl group optionally substituted by 1 to 3 halogen atoms, or (2) a 5- or 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a C_{sub}.1-6 alkyl group optionally substituted by 1 to 3 halogen atoms, and (iii) a C_{sub}.1-6 alkoxy group, and R^{sup.2} is (1) a C_{sub}.6-14 aryl group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, and (ii) a C_{sub}.1-6 alkyl group optionally substituted by 1 to 3 halogen atoms, or (2) a 5- or 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a C_{sub}.1-6 alkyl group optionally substituted by 1 to 3 halogen atoms, and (iii) a C_{sub}.1-6 alkoxy group.

4. The compound or salt according to claim 1, wherein R.^{sup.1} is (1) a C._{sub.3-10} cycloalkyl group, or (2) a 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 halogen atoms, and R.^{sup.2} is (1) a C._{sub.6-14} aryl group optionally substituted by 1 to 3 halogen atoms, or (2) a 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 halogen atoms.
5. A pharmaceutical composition comprising the compound or salt according to claim 1 and a pharmaceutically acceptable carrier.
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